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FUNCTIONAL VERIFICATION TEST REPORT TAS-TOP

PWM and Analogue I/O Module (PAM)

THALES GROUP INTERNAL

Entity Identifier	Document Identifier	DTC	Revision	Customer ID with Revision
15900~9950	7HA-02944-AAAA-980EN	980	07	N/A
Legacy Identifier: *N/A*				

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Template: 83410477-DOC-CAN-EN-009

Follow-up of the evolutions

LOG OF CHANGES				
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02	2012-12-19	CMCS	Michael Sacson	Update
03	2013-10-22	CMCS	Michael Sacson	Update
04	2014-03-19	CMCS	Shawn D'Alimonte	Update
05	2014-10-08	CMCS	Michael Sacson	Update
06	2019-01-21	CMCS	Andy Feng	Update
07P08	2021-03-08	CMCS	Michael Sacson	Traceability update
07	2021-03-08	CMCS	Michael Sacson	Release

APPROVAL		
	Name & Role	
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1 INTRODUCTION

1.1 PURPOSE

The document describes the functional verification of the item Analogue & PWM Main Board (PAM). In the first Edition it specifies the tests. In the following Editions it describes the results of all functional tests, which are carried out during the functional test verification.

The item is part of Thales On-board Platform (TOP) system.

1.2 INTENDED AUDIENCE

The intended audience of this document:

- Development team
- Safety manager (SAF)
- Assessor (AS)
- Validation (VAL)

1.3 EVOLUTION OF THE DOCUMENT

The document is first issued during detailed design phase, and will be updated on feature or requirement changes, if necessary.

For type and reasons of changes please refer to section "Editions and Change History" at the beginning of this document.

1.4 CONFIGURATION MANAGEMENT

This document is under configuration control in CMCS. The document is to be released by the corresponding release authority, defined in the documentation plan /DOP/ of the project. The final version will be archived in PALMA.

1.5 REFERENCES

The mentioned edition is the one used for creation of the document.

This document will only be updated on essential changes of any of its input documents which affect the content of this FVTS/R. Other changes of the input documents and their editions will not lead to a new edition of this FVTS/R.

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1.5.1 Referenced documents

Table 1: Referenced documents

Doc-ID	Document number	ED	Title
[DOP]	7HA-02990-0001-931EN	08	Documentation Plan TOP
[HW-IRS_PAM]	7HA-02922-AAAA-306EN	04	Item Requirement Specification PAM
[CD_PAM]	7HA-02944-AAAA-030EN 7HA-02944-ABAA-030EN 7HA-02944-ACAA-030EN	05 01 02	Circuit Diagram PAM
[HW-ID_PAM]	7HA-02944-AAAA-104EN	06	Hardware Item Description PAM
[EBOM_PAM]	7HA-02944-AAAA-001EN 7HA-02944-BAAA-001EN 7HA-02944-ABAA-001EN 7HA-02944-BBAA-001EN 7HA-02944-ACAA-001EN 7HA-02944-BCAA-001EN 7HA-02944-AAAB-001EN	12 05 03 01 03 01 03	Parts List (EBOM) PAM
[RR-PBA_PAM]	7HA-02944-AAAA-909EN 7HA-02944-BAAA-909EN	04 03	Reliability Report PAM
[IID_PAM]	7HA-02922-AAAA-040EN	05	Assembly Drawing PAM
[INSUCO_TOP]	7HA-02990-0019-959EN	03	Insulation Coordination TOP
[HWADD_TOP]	7HA-02990-0018-434EN	08	HW Architecture Design Document TOP
[PBATD_PAM]	7HA-02944-AAAA-040EN 7HA-02944-BAAA-040EN 7HA-02944-AAAA-400EN 7HA-02944-BAAA-400EN 7HA-02944-AAAA-581EN	04 02 04 01 06	Printed Board Assembly Document Set (Assembly Drawing, PBA Test Data, Design Project File) PAM
[DERATING]	8DC-01043-0001-521EN	01	Component Derating Requirements
[FMDD_TOP]	3CU-04016-0031-549EN	07	TOP - Functional Module Design and Requirements Document

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Doc-ID	Document number	ED	Title
[TEST]	AV-58200-10004-B65A	07	Treatment and test of boards and devices
[QTR-PRD-HW_COMM-IO_SR]	7HA-02950-AAAA-279EN	01	Qualification Test Report Comm-IO Subrack
[ITR_PRD_TOP]	3CU-04016-0052-988EN	08	TOP System Integration Report

1.5.2 Additional references

Table 2: Additional references

Doc-ID	Document number	Title
[Gloss]	8DC-10110-9000-WDAFA	Definitions, and Abbreviations
[SAC_MELF0204]	1AB-13984-0001-905EN	Surface Mount Resistors Metal Film –Size 0204 (MELF) Safety Application Conditions
[SAC_GUIDE]	8AK-00070-0090-955EN	Guidelines and constraints for the usage of inherent fail safe components

1.5.3 Standards

Table 3: Standards

Doc-ID	Publisher	Title
[AREMA]	AREMA	American Railway Engineering and Maintenance of Way Association: Communications and Signals Manual
[EN 50121]	CENELEC	Railway applications – Electromagnetic compatibility
[EN 50124-1]	CENELEC	Railway applications – Insulation coordination – Part 1: Basic requirements – Clearances and creepage distances for all electrical and electronic equipment
[EN 50125-3]	CENELEC	CENELEC: Railway applications - Environmental conditions for equipment - Part 3: Equipment for signalling and telecommunications
[EN 50129]	CENELEC	Railway applications - Communication, signalling and processing systems - Safety related electronic systems for signalling
[EN 50155]	CENELEC	Railway applications – Electronic equipment used on rolling stock

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1.6 DEFINITIONS AND ABBREVIATIONS

1.6.1 Definitions

All definitions used in these documents are determined in [Gloss] except:

Table 4: Document specific definitions

Term	Description
HP (Horizontal pitch)	Measure for printed board front panel with (1 HP = 5.08mm)
U (Unit)	Measure for mounting rack height (1U = 44.45mm)

1.6.2 Abbreviations

All abbreviations used in this document are explained in [Gloss], except:

Table 5: Document specific abbreviations

Abbreviation	Meaning
AREMA	American Railway Engineering and Maintenance of Way Association
ATE	Automated Test Equipment
CAN	Controller Area Network
CB	Check Back (signal)
CQ	ClearQuest Defect tracking System
CoC	Certificate of Compliance
DR	Disconnect Relay
EMC	Electromagnetic Compatibility
ESD	Electro Static Discharge
Fail-safe	A design philosophy applied to safety-critical systems such that the result of hardware failure or the effect of software error shall either prohibit the system from assuming or maintaining an unsafe state, or shall cause the system to assume a state known to be safe
GmbH	Limited (German: Gesellschaft mit beschränkter Haftung)
HU	Hardware Unit
HW	Hardware

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Abbreviation	Meaning
HWR	Hardware Responsible
ID	Identification
I/O	Input / Output
LED	Light Emitting Diode
MTBF	Mean Time Between Failure
MTTR	Mean Time To Repair
NO	Normally Open (contact)
FTC	First Test Cycle
NC	Normally Closed (contact)
PALMA	Product and lifecycle management
PAM	PWM and Analogue Module
PBA	Printed Board Assembly
PCB	Printed Circuit Board
PGA	Programming Gain Amplifier
PVC	Poly Vinyl Chloride
PWM	Pulse Width Modulation
SAC	Safety Application Conditions
SPS	Samples Per Second
SRS	System Requirements Specification
SW	Software
TCIS	Thales Component Information System
TOP	Thales On-board Platform
TVS	Transient Voltage Suppressor
UUT	Unit Under Test
2oo2	Two-out-of-Two System Configuration

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2 SUMMARY

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TEST: LABELLING AND ASSEMBLY 25
Result: Passed 25

TEST CASE: FVTR_OPT_02 25

TEST: FRONT ELEMENTS 25
Result: Passed 25

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TEST: SURFACES 26
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TEST CASE: FVTR_OPT_04 26

TEST: ASSEMBLY CODING 26
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TEST CASE: FVTR_MECH_02 28

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Result: Passed 28

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Result: Passed....33

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TEST: ANALOGUE VOLTAGE-TYPE INPUTS....51
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TEST: ANALOGUE CURRENT-TYPE INPUTS 56
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TEST: PWM OUTPUT FROM PAM BOARD IS NOT CORRECTLY SHUTOFF....87
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TEST: 96V WAS NOT CONSIDERED FOR VARIANT AAAA....89
Result: Passed....89

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3 SUMMARY OF CLEARQUEST TICKETS DURING DESIGN STAGE

Table 6: List of ClearQuest tickets

ID	Severity	Type	Headline	Status
TASCM00234614	3 - Major	Defect	7HA 02944 AAAA PAM Main PBA Update	Solved
TASCM00240677	3 - Major	Defect	7HA 02944 AAAA PAM Main PBA Update	Solved
TASCM00231894	4 - Minor	Improvement	7HA 02922 AAAA PAM: Mechanic	Solved
TASCM00249038	3 - Major	Defect	7HA02944AAAA PAM update as per Bug Fix Cycle functional verification	Solved
TASCM00260142	4 - Minor	Defect	7HA02944xAAA PAM - no test access to the signal EGND from bottom site	Solved
TASCM00236800	-	Improvement	All TOP boards: Rating of fuses	Solved
TASCM00236797	3 - Major	Defect	All TOP I/O boards: Edge Connectors Misalignment	Solved
TASCM00272478	3 - Major	Defect	Connector from Mainboard to Daughterboard to be changed	Solved
TASCM00246198	3 - Major	Defect	PAM Design Generating too much Heat	Solved
TASCM00262317	3 - Major	Defect	Add termination resistor to PAM for PWM output	Rejected
TASCM00269270	3 - Major	Defect	PWM output from PAM board is not correctly shutoff	Rejected
TASCM00307459	4 - Minor	Defect	The components Pos. 1302 (MA302_1 / MA302_1 /) have the false footprint in the ODB-data.	Solved
TASCM00368348	2 - Major	Defect	PAM 7HA-02944-AAAA was designed for 110V power rail and its input voltage range doesn't cover the whole voltage range for 96V, which is stated in the standard EN 50155.	Solved

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4 TEST ASSEMBLY FOR VERIFICATION

4.1 TEST ITEM

Table 7: Test Item

ID	Description	Item number	Ed.	Manufacturing #
MAV1	Double Analog & PWM -Main board	7HA-02944-AAAA	10	EG142910935
MAV2	Double Analog & PWM -Main board	7HA-02944-AAAA	10	EG142910933
MAV3	I/O Microcontroller Daughterboard	7HA-02930-AAAA	11	EG143206030
MAV4	Double Analog & PWM -Main board	7HA-02944-AAAA	07	EG133509402
MAV5	Double Analog & PWM 110V-Main board	7HA-02944-AAAA	15	EG181111835
MAV6	Double Analog & PWM 96V/110V-Main board	7HA-02944-AAAB	02	EG181990465
MAV7	I/O Microcontroller Daughterboard	7HA-02930-AAAB	01	EG174205587
MAV8	Double Analog & PWM 24V/36V-Main board	7HA-02944-ACAA	03	EG182090099
MAV9	Double Analog & PWM 48V/72V-Main board (modify from a PWM 110V board)	7HA-02944-ABAA	09	EG170207610

4.2 TEST CONFIGURATION AND DEFINITIONS

4.2.1 Test environment

Test lab.

4.2.2 Physical test configuration

The setup in Figure 1 was used for most functional tests. Other setups will be described in the specific test descriptions.

Figure 1: Test Set-up

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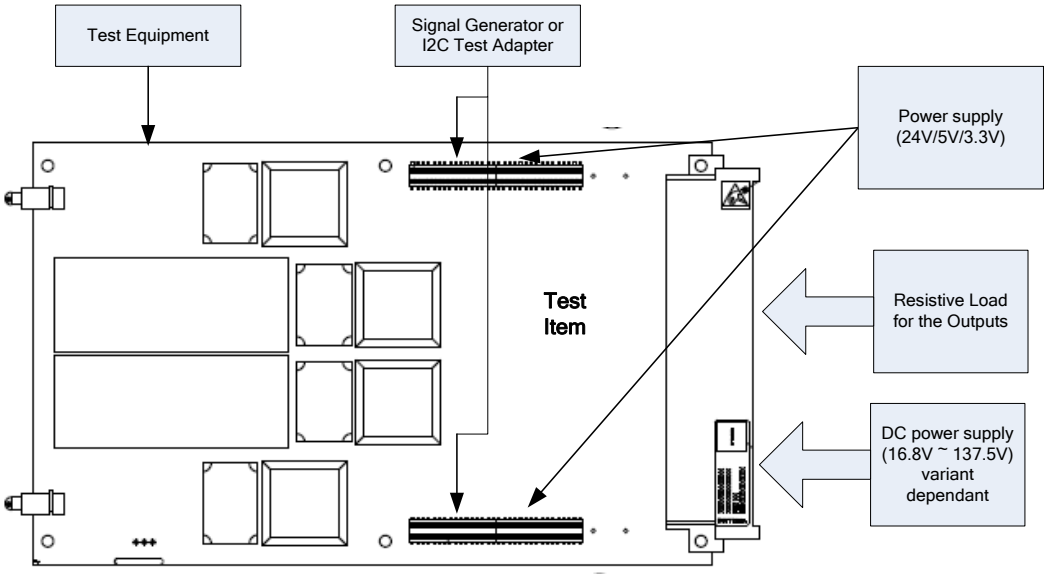

The supply voltages for the board electronics are fed in through the board to board connectors P1 and P2, when the CAN IO daughter board is not equipped.

The PWM Output power supply is fed in the rear connector P3 where the loads are also connected.

By default, the functional tests are normally done without the CAN IO daughter board. If the CAN IO daughter board is necessary for a test then it is mentioned for that test.

4.2.3 Calibrated test equipment

The following calibrated test equipment was used:

Table 8: Calibrated Test Equipment

ID	Equipment	Type	Inventory Number	Calibration Due Date
1	5MHz Function /Arbitrary Generator	33250A	TE0-0374	March 9, 2019
2	Digital Multimeter	FLUKE 289	TE0-0446	Sep. 27,2019
3	Digital Scope	Tektronix TDS2024B +2 P2220 Probes	TE0-0363, T92-0103, T92-0106	March 22, 2019
4	Digital Scope	Infinivision MSOX4054A +4 Probes	TE0-1181	June 28, 2019

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4.2.4 Not calibrated Test Equipment

Table 9: Not calibrated Test Equipment

ID	Device	Type	Inventory Number
1	Power Supply	KIKUSUI PAS160-6	TE0-0543
2	Power Supply	KIKUSUI PAS60-12	T91-0050
3	100Ω Power Rheostat	7141	-
4	100Ω Power Rheostat	7141	-
5	100Ω Power Rheostat	7141	-
6	Test Adapters for connectors P1 and P2	-	-
7	KETEREX USB-910H Embedded System Interface	KXUSB-910H	000390
8	Laptop	Lenovo R61	Thales-10407

4.2.5 Test software

- KETEREX USB-900 Control Center Version 2.1

4.3 INFORMATION CONCERNING THE TESTS

Unless otherwise specified, for the subsequent tests, the following information is valid:

- The signal and component names below refer to the circuit diagram [CD_PAM]

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5 VERIFICATION OF MECHANICAL CHARACTERISTICS

Test case:	**FVTR_OPT_01**
Test:	**Labelling and assembly**
Test Item:	MAV8
Description:	Examination of correct labelling and assembly of parts according to [IID_PAM].
Failure criteria:	Labelling and assembly of parts not correct.
Reference:	[NO_OUT-LINKS: Standard test]
Date:	Aug. 01, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	Labelling and assembly as per [IID_PAM].

[FVTSR_PAM_0002]

Test case:	**FVTR_OPT_02**
Test:	**Front Elements**
Test Item:	MAV8
Description:	Front panel LEDs shall be positioned as described in [HWADD_TOP].
Failure criteria:	LEDs wrongly located.
Date:	Aug. 01, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	Front panel LEDs are positioned as described in [HWADD_TOP].

[HW-IRS_PAM_219]

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Test case:	**FVTR_OPT_03**
Test:	**Surfaces**
Test Item:	MAV8
Description:	Visual inspection of surfaces on non-damaged and proper conditions (bubbles of conformal coating, etc.)
Failure criteria:	Damaged surfaces.
Reference:	[NO_OUT-LINKS: Standard test]
Date:	Aug. 01, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	All surfaces are non-damaged and in proper condition.

[FVTSR_PAM_0004]

Test case:	**FVTR_OPT_04**
Test:	**Assembly coding**
Test Item:	MAV8
Description:	The module shall be coded at the bottom coding block to prevent its insertion into a wrong slot in the subrack, as defined in [HWADD_TOP]. Correct coding shall be match the Table 10.
Test Equipment:	Visual verification
Failure criteria:	Coding pegs missing or incorrect code.
Date:	Aug. 01, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	The coding is correct, as defined in [HWADD_TOP]. For all variants, the assembly coding should match Table 10 below.

Table 10: assembly coding

THALES GROUP INTERNAL

Entity Identifier	Document Identifier	DTC	Revision	Customer ID with Revision
15900~9950	7HA-02944-AAAA-980EN	980	07	N/A
Legacy Identifier: N/A				

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	D	E	F
7HA02944AAAA	3	1	1
7HA02944ABAA	3	1	2
7HA02944ACAA	3	1	3

[HW-IRS_PAM_266]

Test case: **FVTR_OPT_06**

Test: **Assembly drawing**

Test Item: MAV8

Description: The board assembly is realized according to the assembly drawing [PBATD_PAM].

Failure criteria: The board assembly is not complying with assembly drawing.

Reference: [NO_OUT-LINKS: Standard Test]

Date: Aug. 01, 2018

Test carried out by: Andy Feng, HWCC

Test verified by: Michael Sacson, HWCC

Result: **Passed**

Comment: The board assembly is built in accordance with [PBATD_PAM].

[FVTSR_PAM_0008]

Test case: **FVTR_MECH_01**

Test: **Plugging and unplugging of the board assembly**

Test Item: MAV6, MAV7

Description: The board assembly must be pluggable into and extricable from the subrack without excessive force. The board assembly shall be compatible with the TOP standardized I/O slot.

Test equipment: IO Subrack, IO-Expansion Subrack

Failure criteria: Plugging and unplugging could only be done with additional tools.

Date: Aug. 01, 2018

Test carried out by: Andy Feng, HWCC

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Test verified by: Michael Sacson, HWCC

Result: **Passed**

Comment: PAM can be plugged and unplugged in standardized I/O slot without excessive force and additional tools.

[HW-IRS_PAM_93][HW-IRS_PAM_98]

[FVTSR_PAM_0009]

Test case: **FVTR_MECH_02**

Test: **Dimensions**

Test Item: MAV8

Description: The board shall be designed as 100 mm x 160 mm x 6HP (height x depth x width) single EuroCard.

Failure criteria: The board doesn't meet the above described dimensions.

Date: Aug. 01, 2018

Test carried out by: Andy Feng, HWCC

Test verified by: Michael Sacson, HWCC

Result: **Passed**

Comment: The printed board size is 160 x 100 mm, specified in bare board drawing 7HA029330027-003EN. The assembly width is 6HP, fixed by the front plate 3JA690040072, F/N 60 in HU BOM.

[HW-IRS_PAM_94][HW-IRS_PAM_95]

[FVTSR_PAM_0010]

Test case: **FVTR_MECH_03**

Test: **Conformal coating**

Test Item MAV8

Description: The board shall be coated.

Failure criteria: Conformal coating is not present.

Date: Aug. 01, 2018

Test carried out by: Andy Feng, HWCC

Test verified by: Michael Sacson, HWCC

Result: **Passed**

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Comment: *The prototype boards have been ordered as non-coated.*

The test case passed by analysis: drawing 7HA02944AAAA_040EN has information regarding conformal coating. Both EBOM (for PAM Main Board and for CAN IO Daughter Board) includes Conformal Coating Specification 3FG000500001-172DE as a reference.

[HW-IRS_PAM_244]

[FVTSR_PAM_0011]

Test case: **FVTR_MECH_04**

Test: **Thermal Management**

Test Item MAV8

Description: The module shall support operation without forced circulation of air in the subrack.

Failure criteria: The module could not support operation without forced circulation of air in the subrack.

Date: Aug. 01, 2018

Test carried out by: Andy Feng, HWCC

Test verified by: Michael Sacson, HWCC

Result: **Passed**

Comment: The module can operate without forced air circulation of air in the subrack.

[HW-IRS_PAM_102]

[FVTSR_PAM_0012]

Test case: **FVTR_MECH_05**

Test: **Daughter card fits**

Test Item MAV6, MAV7

Description: The CAN I/O daughter card can be mated without excessive force. Sufficient clearance must be maintained between the boards. All mounting screws line up correctly.

Test equipment: CAN I/O daughter card

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Failure criteria:	Boards do not mate easily or components have insufficient clearance.
Date:	Aug. 01, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	The CAN I/O daughter card can be mated without excessive force. Sufficient clearance is maintained between the boards. All mounting screws line up correctly.

[HW-IRS_PAM_53]

6 VERIFICATION OF FUNCTION

Test case:	**FVTR_FKT_01**
Test:	**Wiring**
Test Item	MAV8
Description:	Verification of Wiring of PAM according to circuit diagram [CD-PAM].
Failure criteria	Wiring is not correct.
Reference:	[NO_OUT-LINKS: Standard Test]
Date:	Aug. 01, 2018
Test carried out by:	Arnstadt; Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	Normal flying probe test done by Arnstadt.

[FVTSR_PAM_0014]

Test case:	**FVTR_FKT_02**
Test:	**Layout Constraints**
Description:	The constraints in the CAD layout tool has to be checked

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Verification object:	[PBATD_PAM]
Failure criteria	Constraints not considered
Reference:	[INSUCO_TOP]
Date:	Aug. 01, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	Layout constraints have been reviewed. See Layout constraints documents embedded:

7HA02944AAAA_PAM_WaivedDRC_Error

[HW-IRS_PAM_227][HW-IRS_PAM_114]

[FVTSR_PAM_0015]

Test case:	**FVTR_FKT_03**
Test:	**Component Derating**
Description:	All components have their voltage, power or current stress derated according to [DERATING].
Verification object:	[CD_PAM]
Failure criteria	Components are not selected as per [DERATING] requirements.
Date:	Aug. 01, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	Circuit diagram has been analysed to ensure components are used as per [DERATING] standard requirements.

[HW-IRS_PAM_72]

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[FVTSR_PAM_0016]

Test case:	**FVTR_FKT_04**
Test:	**External interfaces**
Test Item	MAV2
Description:	The pin assignments of all connectors on the board must be realized according to the requirements of the [HW-IRS_PAM]. The signals in the PAM connector with the I/O Processor shall be assigned as per specification given in [HWADD_TOP]. The signal definition between PAM and the I/O Processor board shall be compliant with the specification provided in [HWADD_TOP]. All connections to the module shall be done through the rear edge connectors. The backplane connector shall be DIN 41612 Type F48.
Verification object:	[CD_PAM]
Failure criteria	Pin assignment of verification object is not according to the reference.
Date:	Aug. 01, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	[CD_PAM] has been inspected for compliance with external interfaces.

[HW-IRS_PAM_129][HW-IRS_PAM_269][HW-IRS_PAM_216][HW-IRS_PAM_217][HW-IRS_PAM_96][HW-IRS_PAM_97]

[FVTSR_PAM_0017]

Test case:	**FVTR_FKT_05**
Test:	**PAM Inputs**
Description:	The module shall provide two analogue inputs: one analogue current-type input and one analogue voltage-type input. Each input shall be read by both replicas.

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Verification object:	[CD_PAM], [EBOM_PAM]
Failure criteria	PAM does not provide two analogue inputs as described.
Date:	Aug. 01, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	As per schematic and EBOM reading/review: PAM 7HA02944AAAA,7HA02944AAAB, 7HA02944ABAA and 7HA02944ACAA provides two current-type analogue input and two voltage- analogue type input PAM 7HA02944BAAA provides one current-type analogue input and one voltage- analogue type input Each input provides information to both replicas

[HW-IRS_PAM_55][HW-IRS_PAM_60]

[FVTSR_PAM_0018]

Test case:	**FVTR_FKT_06**
Test:	**PAM outputs**
Description:	The module shall be configurable to provide non-vital isolated outputs to the vehicle propulsion system, as following: - Up to two analogue current-type outputs or analogue voltage-type outputs; - Up to two PWM-type outputs. Default configuration for PAM board variant AAAA, ABAA, ACAA is: - Two analogue current-type outputs; - Two PWM-type outputs (nominal voltage 103V, 60V, 30V accordingly). Default configuration for PAM board variant BAAA, BBAA, BCAA is: - One analogue current-type output; - One PWM-type outputs (nominal voltage 103V, 60V, 30V accordingly). Each analogue/PWM output shall be commanded by one CAN I/O channel (no hardware output voting is required).

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Verification object: [CD_PAM], [EBOM_PAM]

Failure criteria PAM module does not provide the outputs as described above.

Date: Aug. 01, 2018

Test carried out by: Andy Feng, HWCC

Test verified by: Michael Sacson, HWCC

Result: **Passed**

Comment: As per [CD_PAM] analysis:

The PAM board variant AAAA, AAAB, ABAA, ACAA provides:
- Two analogue current-type outputs;

- Two PWM-type outputs (nominal voltage 103V, 60V, 30V accordingly).

The PAM board variant BAAA, BBAA, BCAA provides:
- One analogue current-type output;

- One PWM-type outputs (nominal voltage 103V, 60V, 30V accordingly).

Each analogue/PWM output shall be commanded by one CAN I/O channel (no hardware output voting is required).

All circuitry required for analogue voltage-type outputs are also mounted on the board, but EBOM configuration (via pull-up and pull-down resistors) is required. If voltage-type outputs will be required in future, new PAM variants shall be created.

[HW-IRS_PAM_54][HW-IRS_PAM_58]

[FVTSR_PAM_0019]

Test case: **FVTR_FKT_07**

Test: **Checkback signals monitoring**

Description: The module shall have 2oo2 check-redundant internal architecture.

The checkback signals shall be monitored as follows:

- The checkback signal from one analog output (of one replica) shall be connected to only its own replica for checkback;

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	- The checkback signal from one PWM output (of one replica) shall be connected to only its own replica for checkback; - The checkback signal from each disconnect relay shall be connected to both replicas for checkback.
Verification object:	[CD_PAM]
Failure criteria	Checkback monitoring circuit not implemented as described.
Date:	Aug. 01, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	As per analysis, checkback monitoring circuitry is implemented as required. See also [HW-ID_PAM].

[HW-IRS_PAM_57][HW-IRS_PAM_61][HW-IRS_PAM_62]

[FVTSR_PAM_0020]

Test case:	**FVTR_FKT_08**
Test:	**FAR - type relays disconnect**
Description:	The module shall provide FAR-type disconnect relay(s) for isolating inputs and outputs from external I/Os. Note: Disconnect feature is not needed for analogue voltage-type input for PAM board variant BAAA, therefore a provision for bypass jumper can be made to reduce cost.
Verification object:	[CD_PAM]
Failure criteria	No disconnect function implemented.
Date:	Aug. 01, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	As per [CD_PAM] analysis:

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- The module 7HA02944AAAA, AAAB, ABAA, ACAA is using two FAR-type disconnect relays K1 and K2, with three NO and one NC contacts each.

- The module 7HA02944BAAA, BBAA, BCAA is using one FAR-type disconnect relay K1, with three NO and one NC contacts, and a bypass jumper instead of relay K2.

[HW-IRS_PAM_56]

[FVTSR_PAM_0021]

Test case:	**FVTR_FKT_09**
Test:	**Input Circuitry Power Supplies Monitoring**
Description:	The supply voltages generated on the PAM board and used for input circuits shall be monitored.
Verification object:	[CD_PAM]
Failure criteria	No supply voltages monitoring function implemented.
Date:	Dec. 15, 2020
Test carried out by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	As per [CD_PAM] analysis:

- The PAM HU has one Voltage Input and one Current Input, replicated to Replica 1 and Replica 2. In every Replica 4-input ADC MX1 (MX2) is converting the scaled voltage and current values to serial I2C format. To avoid common point failure, input circuitry for Replica 1 and Replica 2 are supplied from two independent DC/DC converters MA1 (MA2), generating voltages P5V_ISO1 (P5V_ISO2).

- One of the channels ADC MX1 (MX2) is used for P5V_ISO1 (P5V_ISO2) voltage monitoring. The measured voltages in I2C format will be transmitted via bi-directional I2C Digital Isolator type ADuM2250 MN3 (MN4) to Replica Connector P1 (P2).

[HW-IRS_PAM_239]

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[FVTSR_PAM_0022]

Test case:	**FVTR_FKT_10**
Test:	**FAR disconnect relays control**
Description:	Each disconnect relay shall be commanded by two CAN I/O channels in the "AND" configuration (2oo2 hardware voting is required for disconnect relays). It translates to a digital signal from the neighbouring replica to control the disconnect relay.
Verification object:	[CD-PAM]
Failure criteria	Disconnect relays are not controlled according to above description.
Date:	Aug. 01, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	As per [CD_PAM] analysis, the disconnect relays are controlled as required. See also [HW-ID_PAM].

[HW-IRS_PAM_59]

[FVTSR_PAM_0023]

Test case:	**FVTR_FKT_11**
Test:	**Variant design consideration for the PAM PCB**
Description:	There shall be only one PCB variant, which includes: -Two PWM Outputs (configurable to required output voltage by BOM change); -Two Current Outputs (configurable as Voltage Outputs by BOM change); -One Current Input; -One Voltage Input.
Verification object:	[CD-PAM], [PBATD_PAM].
Failure criteria	The PAM PCB does not meet the above variant design requirements.
Date:	Aug. 01, 2018

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Test carried out by: Andy Feng, HWCC

Test verified by: Michael Sacson, HWCC

Result: **Passed**

Comment: The PAM PCB design has all provisions to be used in different PAM configurations without PCB updating. Both existing variants are using the same PCB.

[HW-IRS_PAM_73]

[FVTSR_PAM_0024]

Test case: **FVTR_FKT_12**

Test: **Analogue voltage-type outputs**

Description: The PAM board shall support analogue voltage-type outputs with the following characteristics:

Maximum load current: 10 mA;

Output voltage range: 0 to10 VDC;
(Derived from 24 V internal power supply or optional from external power supply);

Minimum resolution: 8 bits;

Verification object: [CD-PAM]

Failure criteria Analogue voltage-type outputs do not meet the above requirements.

Date: August 18, 2014

Test carried out by: Michael Schueler, HWCC

Result: **Passed**

Comment: Analogue voltage-type outputs meet the above requirements.

[HW-IRS_PAM_75]

[FVTSR_PAM_0025]

Test case: **FVTR_FKT_13**

Test: **Analogue current-type outputs**

Test Item MAV1

Description: The PAM board shall support analogue current-type outputs with the following characteristics:

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- Current range: 0 to 24mA;
- Maximum load: 800R with internal DC/DC power supply;
- Minimum resolution: 8 bits;

Failure criteria Analogue current-type outputs do not meet the above requirements.

Date: August 22, 2014

Test carried out by: Michael Schueler, HWCC

Result: **Passed**

Comment: The test has been performed by writing command and reading checkback via I2C bus using KETEREX USB-910H Embedded System Interface. The calculated values have been compared with actual output current, measured by Ammeter.

The test results for Analogue Current Output are provided for both Outputs 1 and 2 in the Table **11** below:

Table 11: Analogue Current Output Test Results

DAC I2C [HEX[	*CURRENT OUTPUT EXPECTED*	*OUT #*	*ADC CHECKBACK [STATUS]*				*CURRENT OUTPUT MEASURED*			
			I2C ZERO LOAD		*I2C [HEX/mA] 800R LOAD*		*I_OUT @ ZERO LOAD [mA]*	*ACC. [%]*	*I_OUT @ 800Ω LOAD* [mA]	*ACC. [%]*
			[HEX]	*[mA]*	*[HEX]*	*[mA]*				
000	0.000	1	001D	0.023	001D	0.023	0.036	N/A	0.036	N/A
		2	0020	0.025	0020	0.025	0.038	N/A	0.038	N/A
1F4	2.930	1	0E9E	2.924	0E9E	2.924	2.943	-0.65	2.943	-0.65
		2	0E93	2.915	0E93	2.915	2.932	-0.58	2.932	-0.58
5DC	8.791	1	2BDE	8.774	2BDE	8.774	8.804	-0.34	8.804	-0.34
		2	2BCA	8.758	2BCA	8.758	8.783	-0.28	8.783	-0.28
9C4	14.652	1	4919	14.620	4918	14.619	14.663	-0.29	14.663	-0.30
		2	490A	14.608	4909	14.607	14.642	-0.23	14.641	-0.23
DAC	20.513	1	664D	20.461	664B	20.459	20.459	+0.01	20.514	-0.27

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		2	6634	20.441	6632	20.440	20.483	-0.21	20.482	-0.21
FFF	24.000	1	77B0	23.938	77AB	23.934	24.060	-0.51	23.998	-0.27
		2	778D	23.911	7789	23.908	23.908	+0.01	23.955	-0.20

Formula for ADC Checkback Status via I2C [mA] calculation (Analogue Current Status):

I2C [mA] = (256[mV]/32767)*(I2C READING)/10[Ω], where:

- 256[mV] = 4096/16 is full scale range for ADC, where 16 is ADC PGA;
- (256[mV]/32767) is the ADC resolution;
- (I2C READING) is the decimal value of *I2C [HEX] READING;*
- 10[Ω] is the value of current sensing resistor R432.

Resolution of Digital-to-Analogue converter and Analogue Output Driver is 12 bit.

Resolution of ADC reading Output Checkback Status is 16 bit.

The accuracy of generated analogue current across the whole range for both Replicas at maximum load 800Ω for the variant tested (with internal DC/DC converter) is better than 1.5%.

[HW-IRS_PAM_76]

[FVTSR_PAM_0026]

Test case: **FVTR_FKT_14**

Test: **PWM outputs**

Test Item **MAV1,** MAV8, MAV9

Description: The PAM board shall support PWM outputs with the following characteristics:

Configurable frequency: from 400Hz to 2500Hz with 10Hz increment;

Voltage: 30VDC, 60VDC, 103VDC nominal (-30%/+25%) range derived from external source;

Maximum load current: 560mA;

Variable (software controlled) duty cycle: from 5% to 95%;

Failure criteria PWM Outputs do not met the above requirements.

Date: MAV1: Sep.3, 2014; MAV8:Aug.02,2018; MAV9: Dec.11,2018

Test carried out by: MAV1: Michael Sacson; MAV8: Andy Feng; MAV9:Andy Feng

Test verified by: Michael Sacson, HWCC

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Result: **Passed**

Comment:

1. The prototype 7HA02944AAAA,7HA02944AAAB for nominal voltage 103 VDC has been tested using PWM signal generated by Signal Generator and met the requirements, see pictures from (fig 2—fig6) ; Test board: MAV1

2. The prototype 7HA02944ACAA for nominal voltage 30 VDC has been tested using PWM signal generated by Signal Generator and met the requirements, see pictures from (fig 7—fig10); Test board:MAV8

3. The prototype 7HA02944ABAA for nominal voltage 60 VDC has been tested using PWM signal generated by Signal Generator and met the requirements, see pictures from (fig 11—fig14); Test board: MAV9

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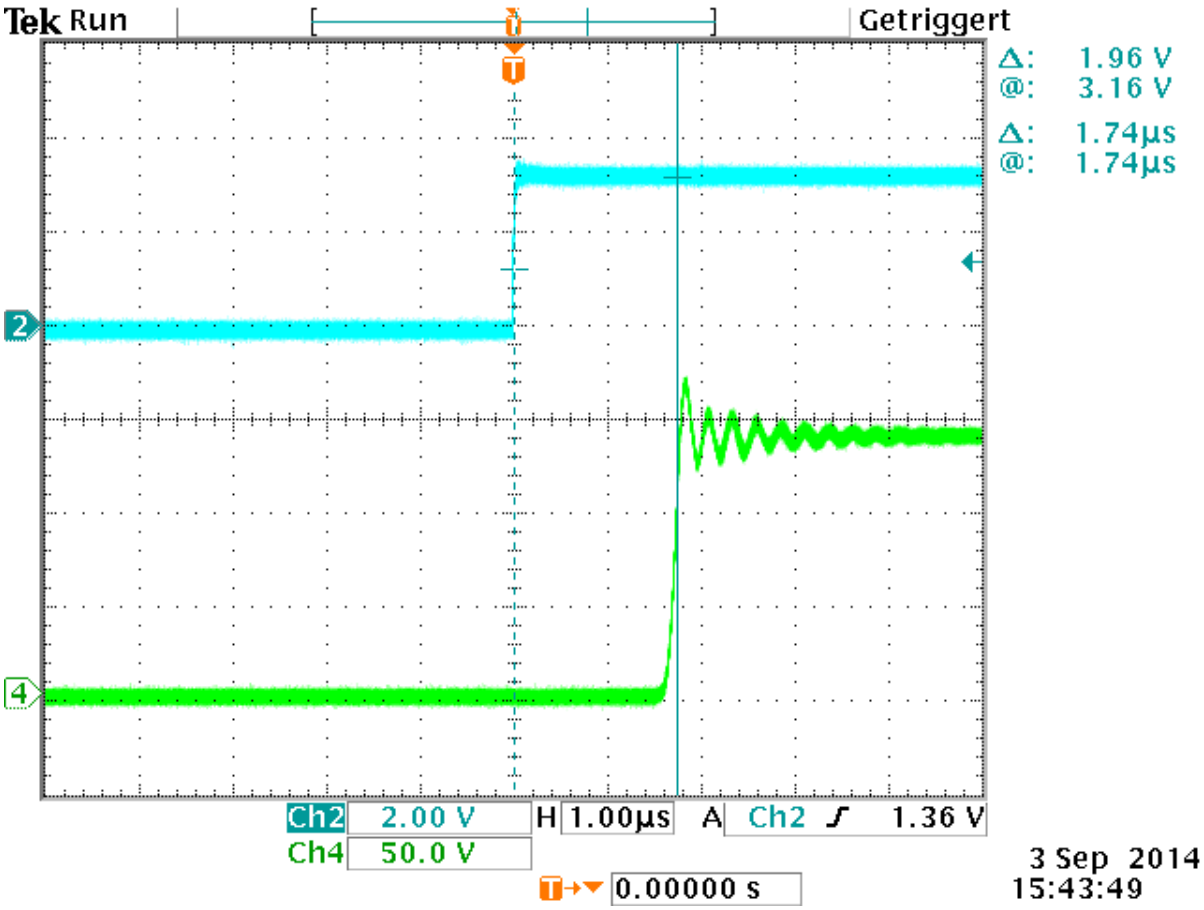

Figure 2: PWM Output Turn-on Delay

CH2 – PWM control signal from generator; CH4 – PWM1 Output under 560 mA load

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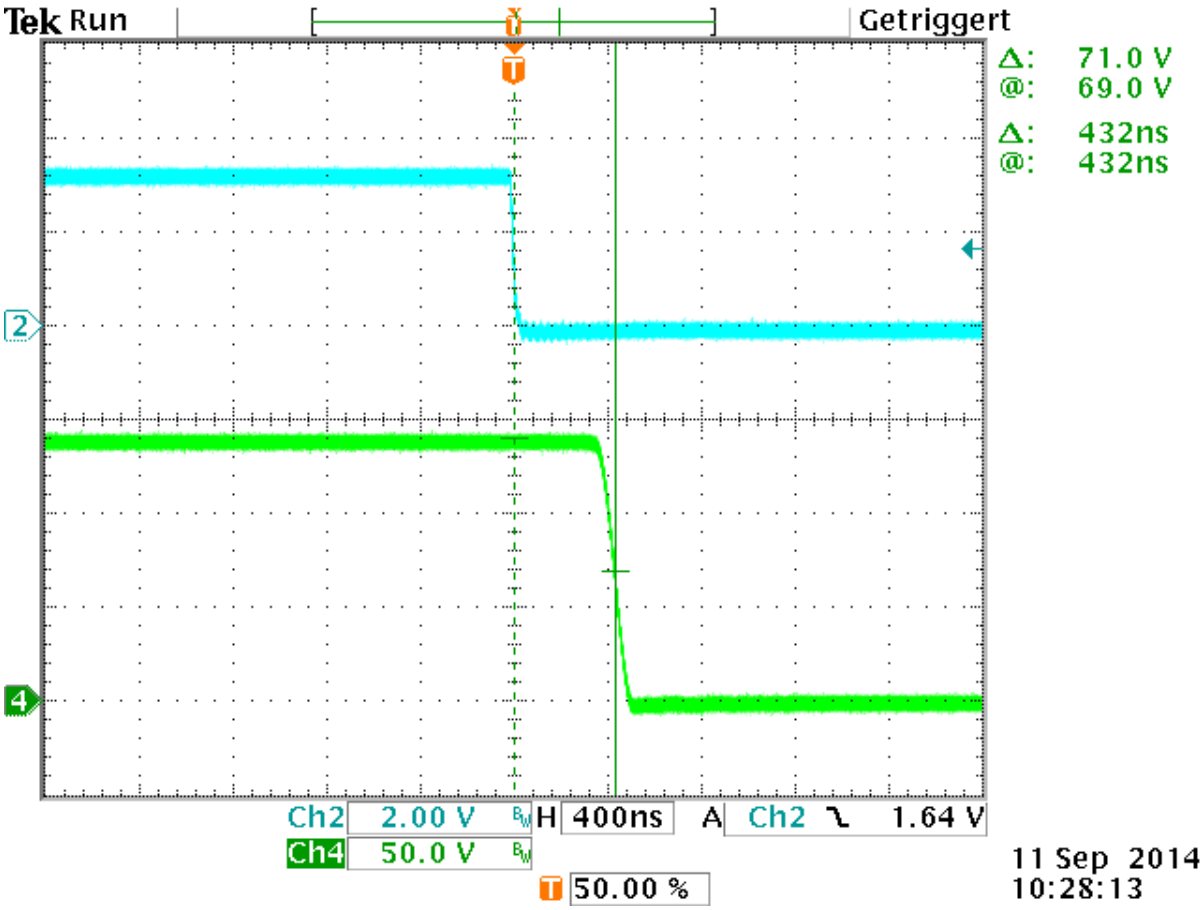

Figure 3: PWM Output Turn-off Delay

CH2 – PWM control signal from generator; CH4 – PWM Output under 140V/560 mA load

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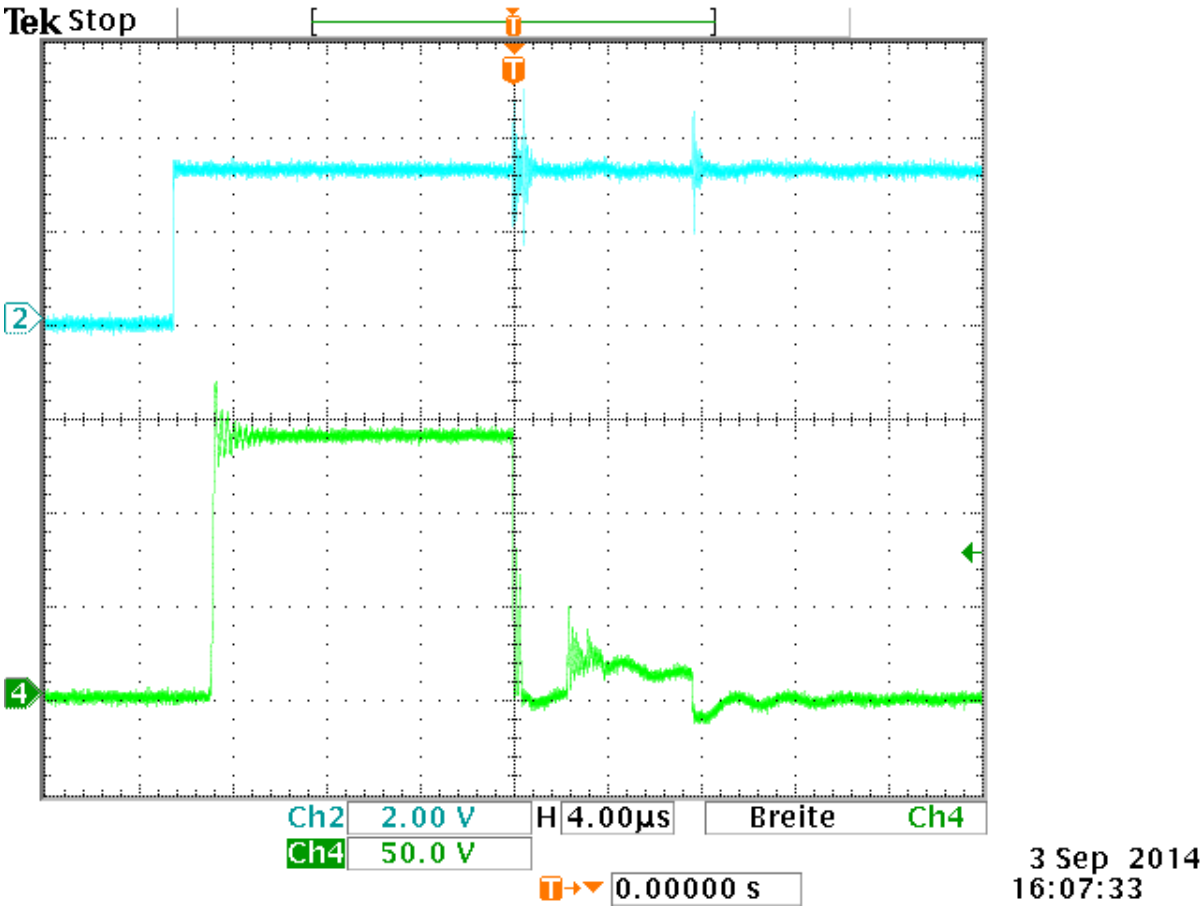

Figure 4: PWM Output in Short Circuit Protection Mode

CH2 – PWM control signal from generator; CH4 – PWM Output shorted at 140V/ 560 mA load

Due to connection of GND for CH2 and CH4 the signal from the generator looks distorted during short circuit test.

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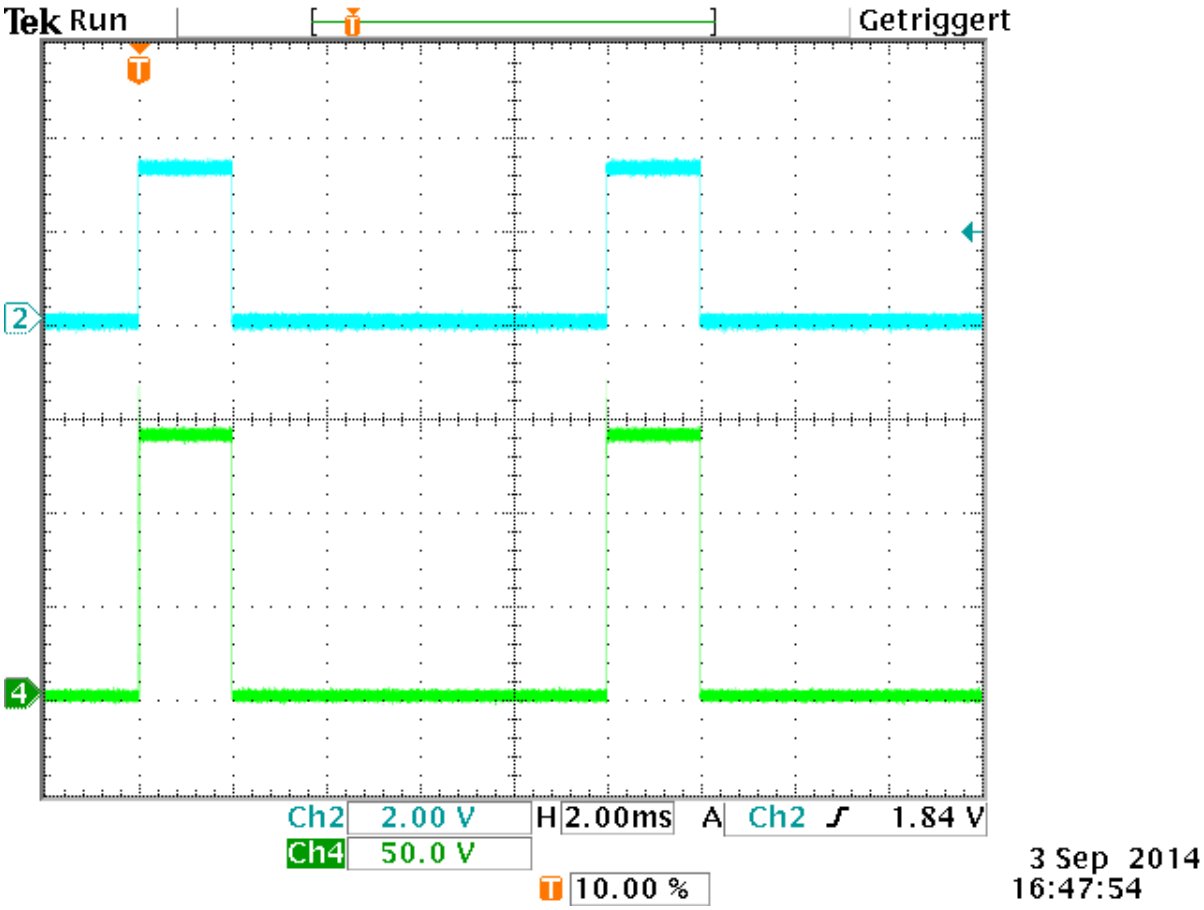

Figure 5: Typical PWM Output at 100Hz

CH2 – PWM control signal from generator; CH4 – PWM Output at 140V/ 560 mA load

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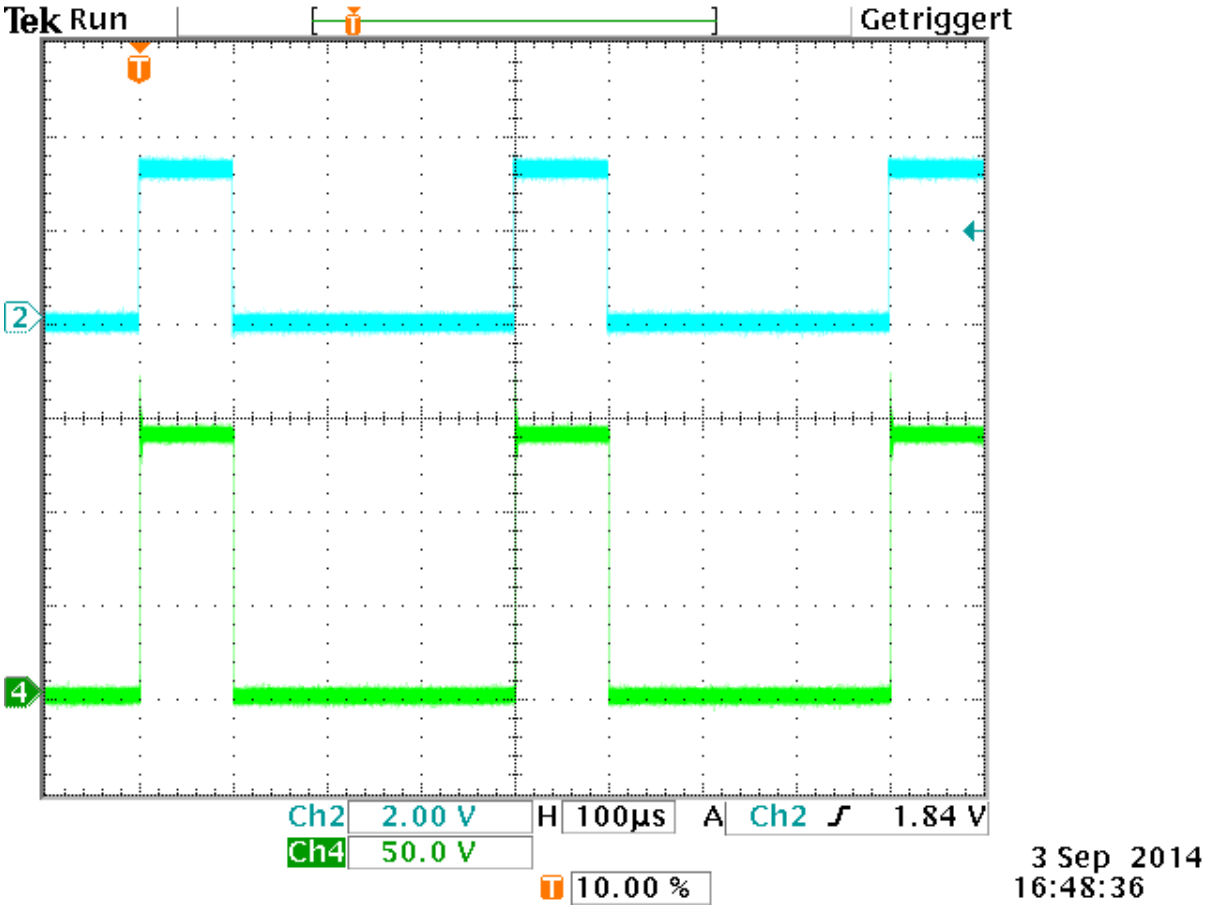

Figure 6: Typical PWM Output at 2500Hz

CH2 – PWM control signal from generator; CH4 – PWM Output at 140V/ 560 mA load.

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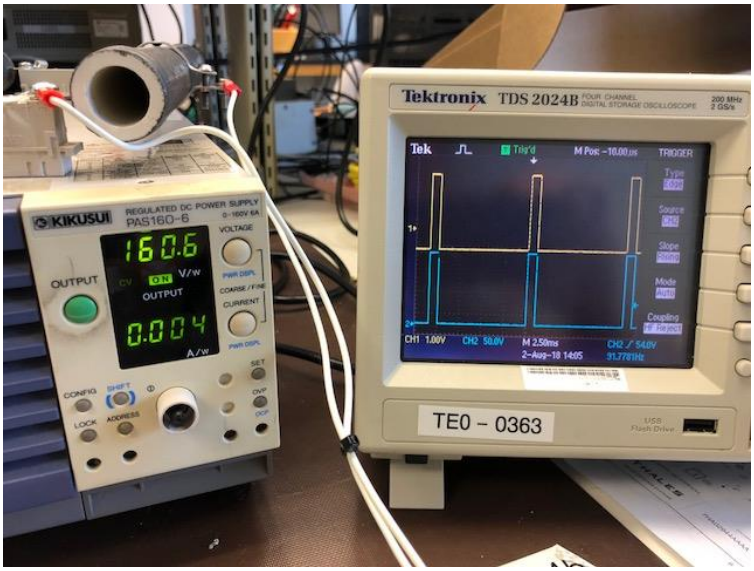

Figure 7: Typical AAAB Variant PWM Output at 100Hz,duty cycle 10%

CH1 – PWM control signal from generator; CH2 – PWM Output at 160V/ 560 mA load.

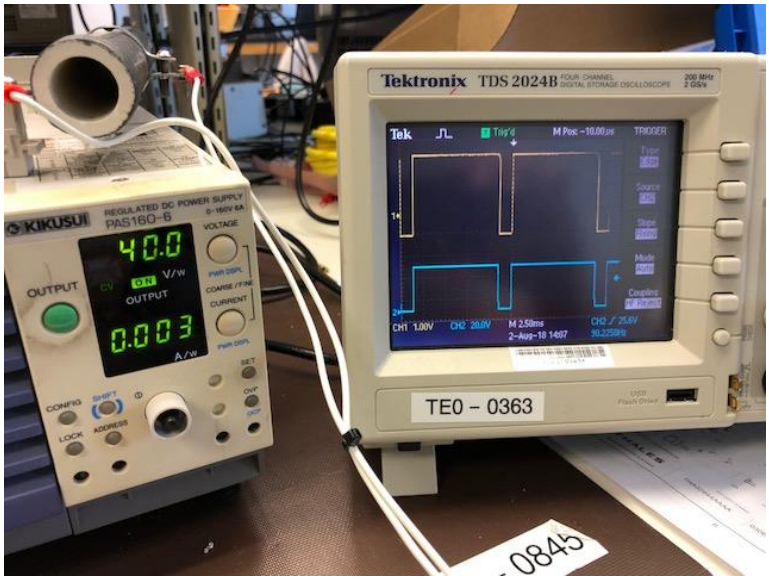

Figure 8: Typical AAAB Variant PWM Output at 100Hz, duty cycle 10%

CH1 – PWM control signal from generator; CH2 – PWM Output at 40V/ 560 mA load.

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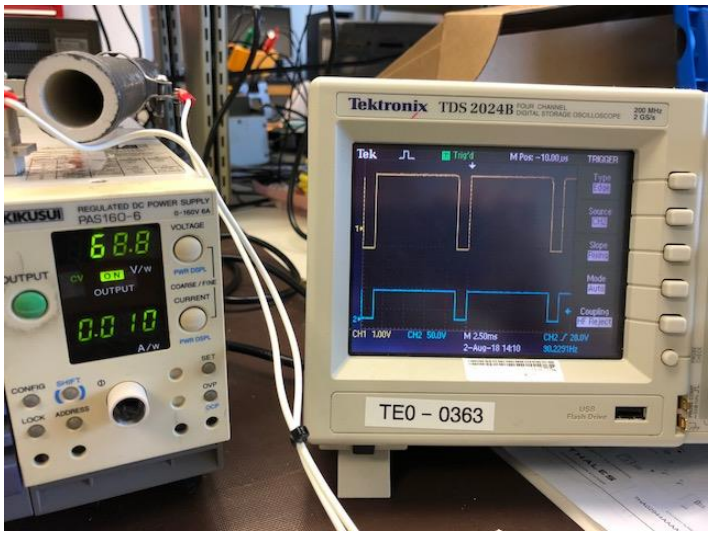

Figure 9: Typical ACAA Variant PWM Output at 100Hz, duty cycle 10%

CH1 – PWM control signal from generator; CH2 – PWM Output at 68V/ 560 mA load.

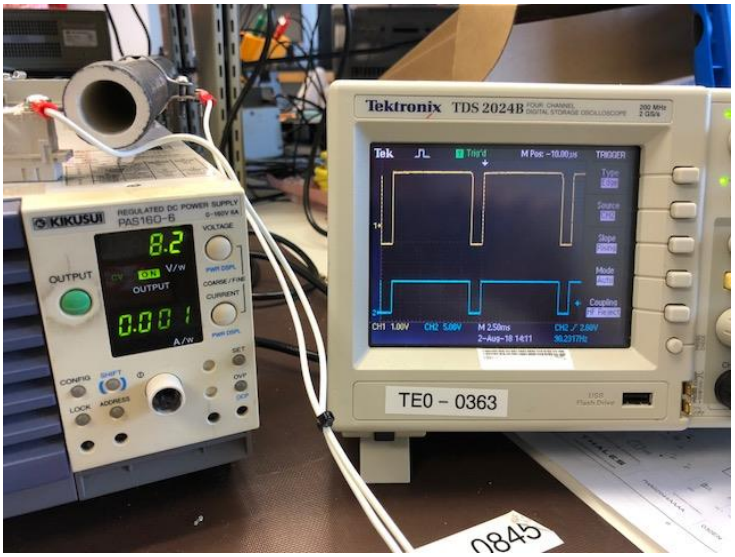

Figure 10: Typical ACAA Variant PWM Output at 100Hz, duty cycle 10%

CH1 – PWM control signal from generator; CH2 – PWM Output at 8.2V/ 560 mA load.

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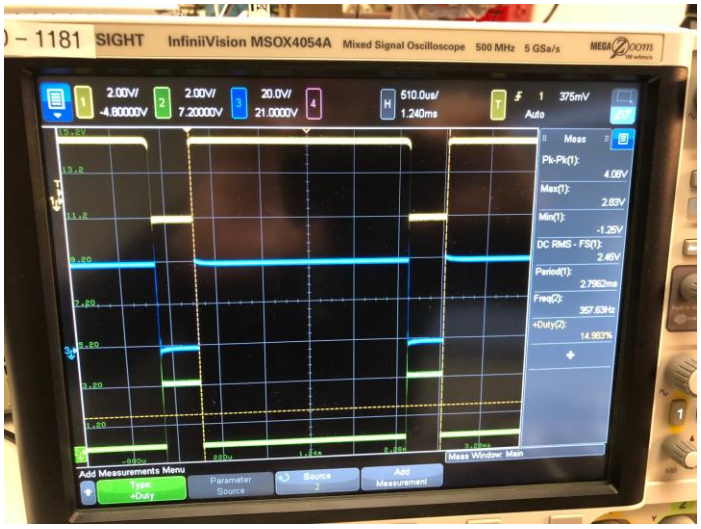

Figure 11: Typical ABAA Variant PWM Output at 357Hz, duty cycle 15%

CH1 – PWM control signal from generator; CH3 – PWM Output at 40V/ 560 mA load; CH2 – PWM Output check back signal;

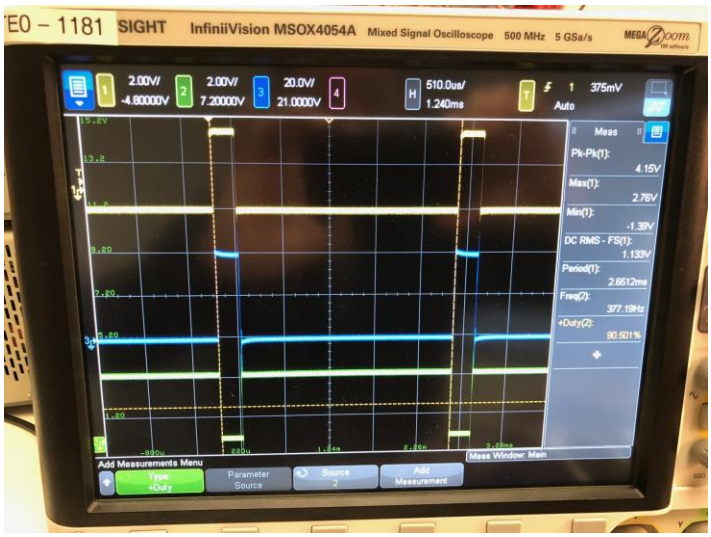

Figure 12: Typical ABAA Variant PWM Output at 377Hz, duty cycle 90%

CH1 – PWM control signal from generator; CH3 – PWM Output at 40V/ 560 mA load; CH2 – PWM Output check back signal.

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Figure 13: Typical ABAA Variant PWM Output at 3.8KHz, duty cycle 90%

CH1 – PWM control signal from generator; CH3 – PWM Output at 40V/ 560 mA load; CH2 – PWM Output check back signal;

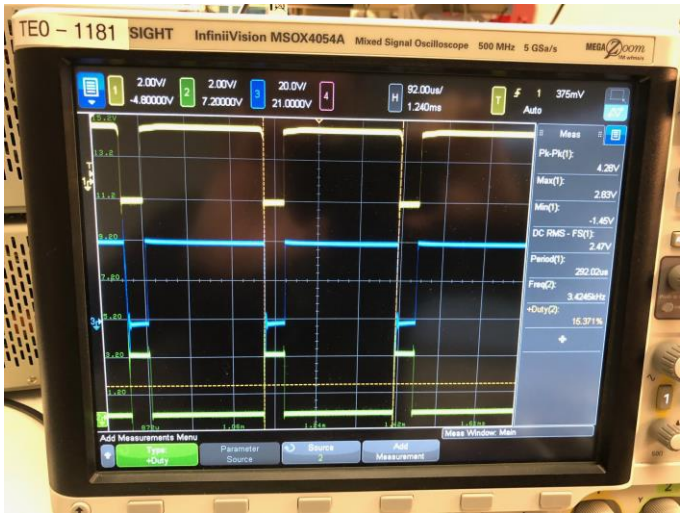

Figure 14: Typical ABAA Variant PWM Output at 3.4KHz, duty cycle 15%

CH1 – PWM control signal from generator; CH3 – PWM Output at 40V/ 560 mA load; CH2 – PWM Output check back signal;

[HW-IRS_PAM_77]

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[FVTSR_PAM_0027]

Test case:	**FVTR_FKT_15**
Test:	**Analogue voltage-type inputs**
Test Item	MAV1
Description:	The PAM board shall support analogue voltage inputs with noise filtering having the following characteristics: Voltage range of 0 to 20VDC with minimum 10kΩ input resistance; Minimum resolution: 10 bits; Minimum cut-off frequency: 5 Hz; Accuracy digitized data: < 2 %; Noise filtering: attenuation for signal with frequency < 5Hz: max. 0.2dB; Protection against transient and over-voltage > 28V (example transient voltage suppressor SMBJ28CA); Protection against reverse polarity; Conversion to the digital format at a rate higher than 35 Hz (7 times of the highest input signal frequency);
Failure criteria	Analogue voltage-type inputs do not met the above requirements.
Date:	September 4, 2014, September 11, 2014
Test carried out by:	Michael Schueler, HWCC
Result:	**Passed**
Comment:	The test has been performed by supplying input voltage and reading ADC output via I2C bus using KETEREX USB-910H Embedded System Interface. The calculated values have been compared with actual input voltage, measured by Voltmeter.

1. The Test results for the analogue input voltage measurement for Replica 1 are shown in the Table below:

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Table 12: Replica 1 Analogue Voltage Input Test Results

V_IN [V]	*V_IN_R1* [V] *@ TB70/GND_IN1*	*I2C [HEX] READING*	*I2C [V] CALCULATED*	*ACCURACY [%]*
0.000	0.0018	0036	0.034	N/A
2.000	0.3970	0C45	1.972	-1.40
4.000	0.7935	18CD	3.987	-0.33
6.000	1.1901	2507	5.952	-0.80
8.000	1.5865	3127	7.902	-1.23
10.000	1.9829	3DB0	9.917	-0.83
12.000	2.3802	49AC	11.843	-1.31
14.000	2.7760	5664	13.888	-0.80
16.000	3.1728	6293	15.846	-0.96
18.000	3.5690	6EFC	17.841	-0.88
20.000	3.9657	7AD1	19.744	-1.28
V_TEST_R1	3.7400	7450	3.722	+0.86

Formula for I2C [V] calculation (Analogue Input Voltage):

I2C [V] = (4.096/32767)*((27.4+6.81)/6.81)*(I2C READING), where:

- (4.096/32767) is the ADC resolution;
- ((27.4+6.81)/6.81) is input voltage dividing coefficient, resistors 27k4 and 6k81;
- (I2C READING) is the decimal value of *I2C [HEX] READING.*

Formula for I2C [V] calculation (V_TEST_R1 voltage):

I2C [V] = (4.096/32767)*(I2C READING)

2. Test results for the analogue input voltage measurement for Replica 2 taken during FTC are shown below:

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Table 13: Replica 2 Analogue Voltage Input Test Results

V_IN [V]	*V_IN_R2* [V] *@ TB72/GND_IN2*	*I2C [HEX] READING*	*I2C [V] CALCULATED*	*ACCURACY [%]*
0.000	0.0254	0027	0.024	N/A
2.000	0.3968	0C36	1.963	-1.85
4.000	0.7937	1884	3.941	-1.48
6.000	1.1905	2542	5.989	-0.18
8.000	1.5874	3195	7.971	-0.36
10.000	1.9844	3DCC	9.934	-0.66
12.000	2.3814	49C7	11.860	-1.17
14.000	2.7776	5643	13.871	-0.92
16.000	3.1746	62A0	15.855	-0.91
18.000	3.5714	6EE9	17.830	-0.94
20.000	3.9689	7B72	19.845	-0.78
V_TEST_R2	3.7500	721E	3.652	-2.61

Formulas for I2C [V] calculation are the same as for Replica 1.

3. See scope screenshots for signal attenuation measurement:

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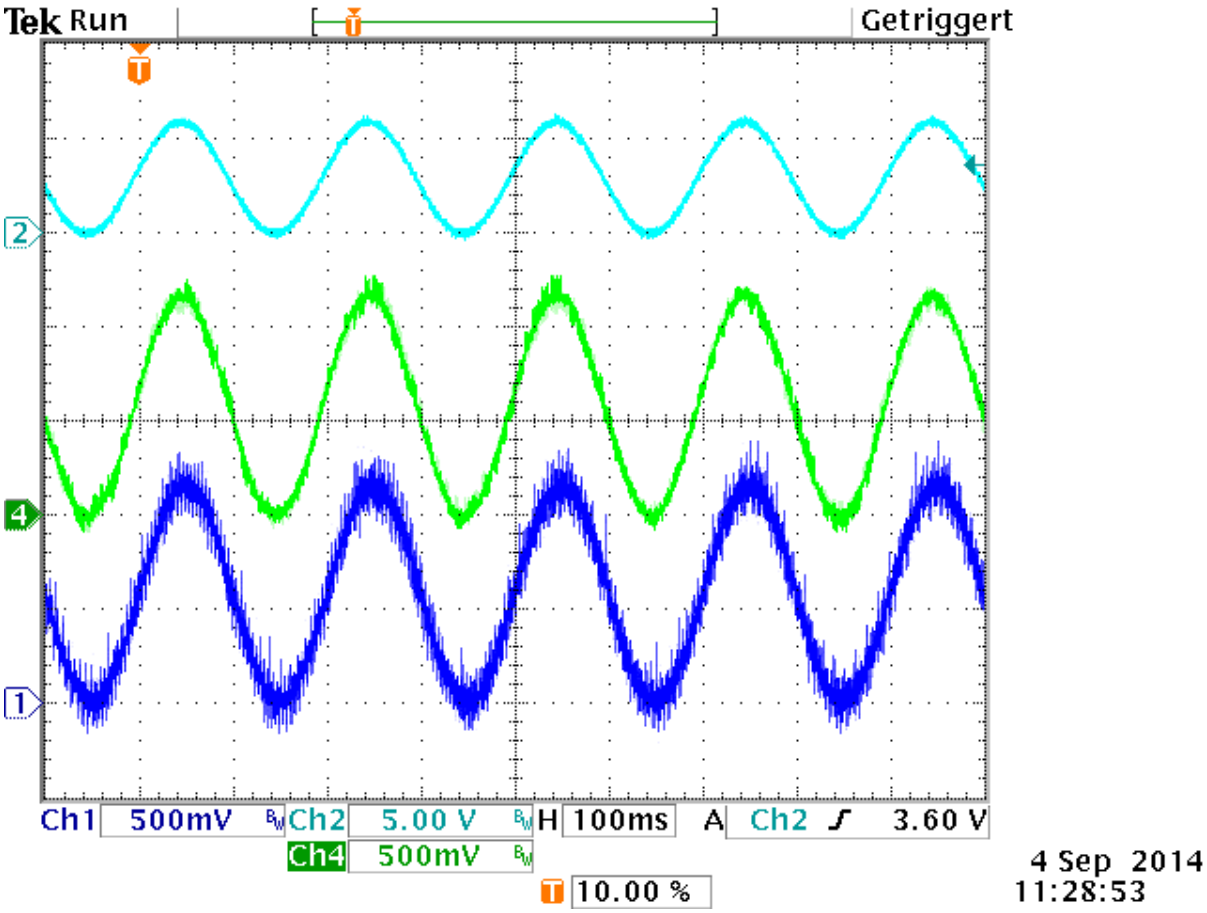

Figure 15: Input Voltage Signal Attenuation Measurement at 5 Hz

CH2 – Input signal from generator at TB70; CH1 – Signal after filter at TB83.

No signal attenuation detected at 5 Hz.

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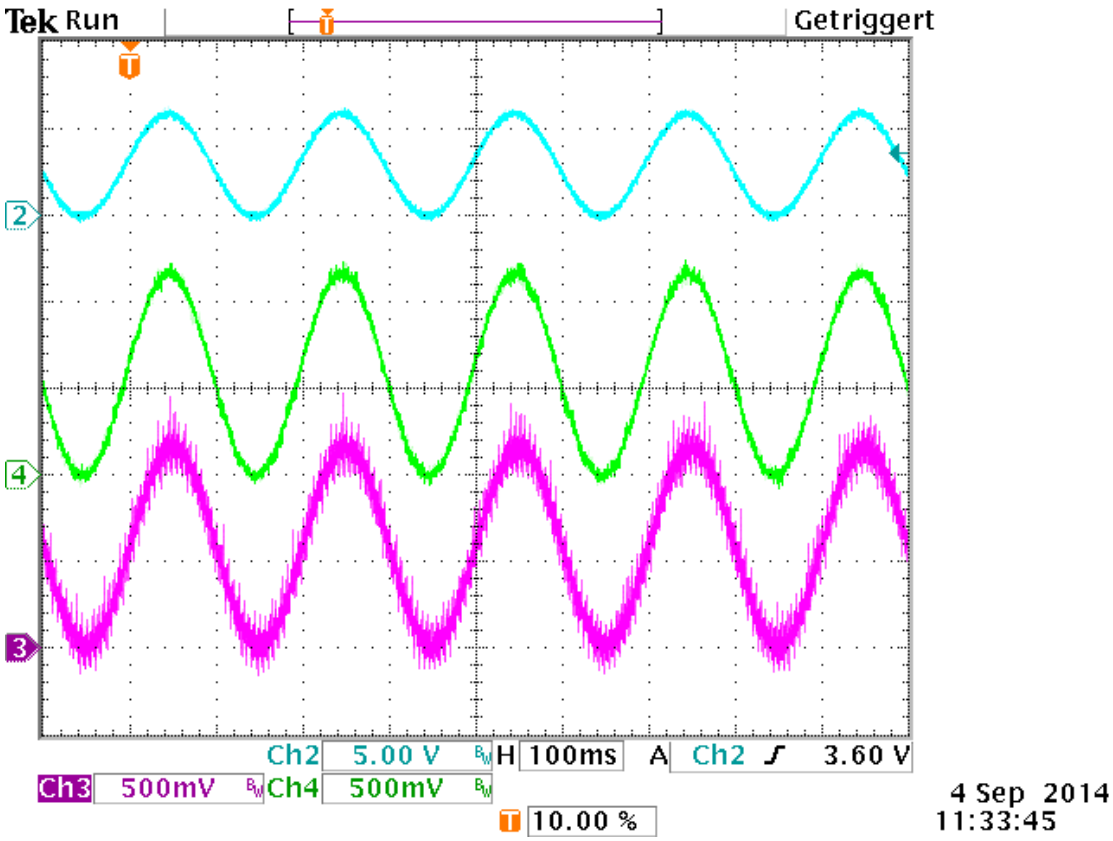

Figure 16: Input Voltage Signal Attenuation Measurement at 50 Hz

CH1 – Input signal from generator at TB70; CH1 – Signal after filter at TB83.

5. The Analogue Input circuit has the following parameters:

- Input Resistance 17k Ω;
- ADC Resolution 16 bit;
- Calculated cut-off frequency 26Hz;
- Accuracy digitized data <1.1 %;
- Transient and reversed polarity protection by SMBJ20A;
- ADC Conversion Rate programmable from 8 to 860 SPS, as tested at 128 SPS;

[HW-IRS_PAM_78]

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[FVTSR_PAM_0028]

Test case:	**FVTR_FKT_16**
Test:	**Analogue Current-type inputs**
Test Item	MAV1
Description:	The PAM board shall support analogue current inputs with the following characteristics: 0 to 20mA input current with maximum shunt resistance of 500Ω. Minimum resolution: 10 bits; Accuracy digitized data: < 2 %; Noise filtering: attenuation for signal with frequency < 5Hz: max. 0.2dB Protection against reverse polarity and over-current > 100 mA Conversion to the digital format at a rate higher than 35 Hz (7 times of the highest input signal frequency)];
Failure criteria	Analogue current-type inputs do not met the above requirements.
Date:	September 4, 2014
Test carried out by:	Michael Schueler, HWCC
Result:	**Passed**
Comment:	The test has been performed by supplying input current and reading ADC output via I2C bus using KETEREX USB-910H Embedded System Interface. The calculated values have been compared with actual input current, measured by Ammeter.

1. Test results for the input current range and accuracy taken are shown in the Table below:

Table 14: Analogue Current Input Test Results

THALES GROUP INTERNAL

Entity Identifier	Document Identifier	DTC	Revision	Customer ID with Revision
15900~9950	7HA-02944-AAAA-980EN	980	07	N/A
Legacy Identifier: N/A				

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INPUT CURRENT MEASURED	*REPLICA CURRENT MEASURED [HEX] AND CALCULATED [mA]*				*INPUT CURRENT CALCULATED*	
I_IN, [mA]	*I_IN_R1*		*I_IN_R2*		*I_IN_R1+I_IN_R2 [mA]*	*ACCURACY [%]*
	[HEX]	*[mA]*	*[HEX]*	*[mA]*		
0.000	0074	0.037	0032	0.016	0.053	N/A
2.000	0C42	1.001	0C42	1.001	2.002	0.10
4.000	1871	1.995	1873	1.996	3.968	-0.22
6.000	2471	2.975	248F	2.984	5.948	-0.68
8.000	30CE	3.984	3042	3.940	7.932	-0.95
10.000	3CCC	4.963	3CBF	4.959	9.890	-0.78
12.000	48DC	5.948	48BC	5.938	11.874	-0.95
14.000	54DB	6.927	54DE	6.928	13.853	-1.04
16.000	61DF	7.990	60BF	7.898	15.813	-0.70
18.000	6D6A	8.932	6D45	8.920	17.796	-0.82
20.000	7968	9.911	7944	9.900	19.777	-0.95

Formula for ADC Replica Current via I2C [mA] calculation (Analogue Current Status):

I_IN_Rx [mA] = (4096[mV]/32767)*(I2C READING)/392[Ω], where:

- (4096[mV]/32767) is the ADC resolution;
- (I2C READING) is the decimal value of *I_IN_Rx [HEX] READING;*
- 392[Ω] is the value of current sensing resistor R26 (R28);

2. The Analogue Input circuit has the following parameters:

- Input Resistance 477 Ω;
- ADC Resolution 16 bit;
- Calculated cut-off frequency 26Hz;
- Accuracy digitized data <1.2 %;
- Transient and reversed polarity protection by SMBJ10A;

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- Short circuit protection by fuse 0.5A;
- ADC Conversion Rate programmable from 8 to 860 SPS, as tested at 128 SPS.

3. Input Current is split equally between Replica 1 and Replica 2. Actual measured value for Input Current shall be calculated as a sum of measurements by Replica 1 and Replica 2, see column *INPUT CURRENT CALCULATED* in the Table above.

4. See scope screenshot for signal attenuation measurement:

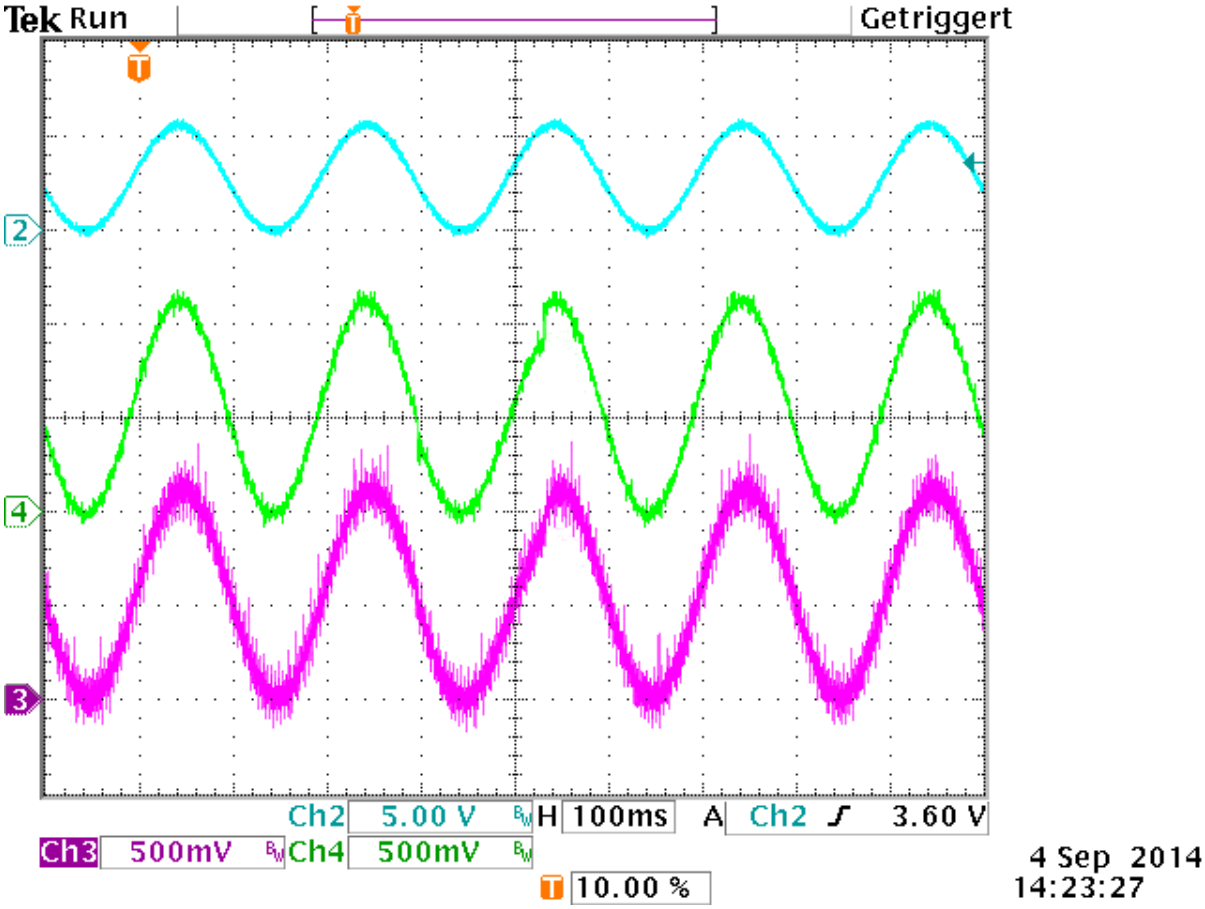

Figure 17: Input Current Signal Attenuation Measurement at 5 Hz

CH1 – Input signal from generator at TB69; CH2 – Signal after filter at TB85

No signal attenuation detected at 5 Hz.

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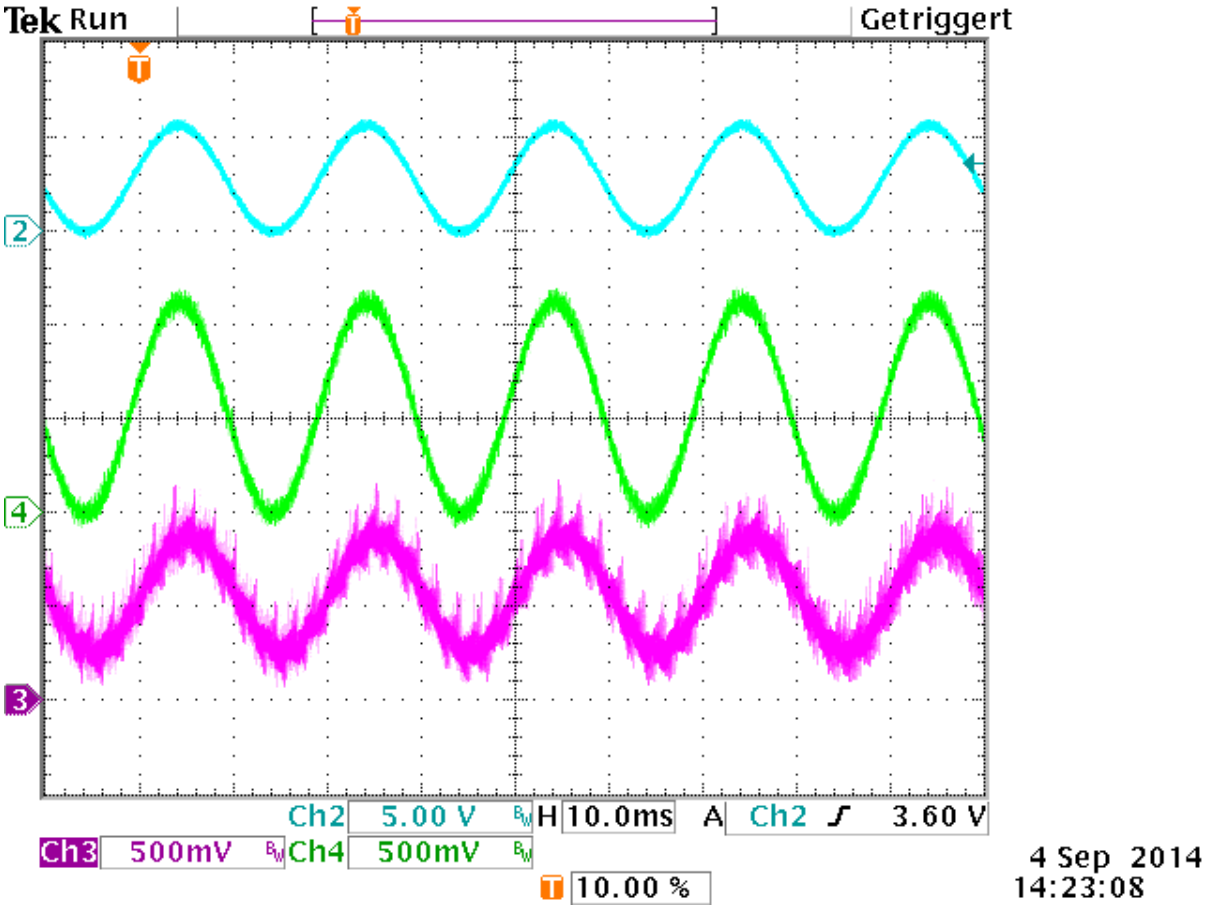

Figure 18: Input Current Signal Attenuation Measurement at 50 Hz

CH2 – Input signal from generator at TB69; CH2 – Signal after filter at TB85.

[HW-IRS_PAM_79]

[FVTSR_PAM_0029]

Test case:	**FVTR_FKT_17**
Test:	**Grounding and provisions for electrostatic discharge at insertion**
Test Item	MAV8
Description:	The PAM board shall use common signal ground configuration (no separate ground for replicas). To reduce the effects of

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electrostatic discharge during the board insertion, discharge strips will be provided along the edges of the board.

The ESD strip on the printed board consists of two separate segments, which are not connected with each other. One segment reaches the front plate; the other one reaches the rear end of the board and is connected with electronic ground via a resistor. The gap between the segments is larger than the contact slider of the subrack.

Failure criteria Grounds of the replicas are separated or discharge strips have not been implemented.

Date: Aug. 02, 2018

Test carried out by: Andy Feng, HWCC

Test verified by: Michael Sacson, HWCC

Result: **Passed**

Comment: The ESD Strip is implemented according to information provided in [HW-IRS_PAM_83] picture;

Resistance front plate – front-plate segment <0.1Ω, resistor rear-end segment – GND 536kΩ. The gap between the segments is 12mm, much larger than slider contact surface.

Grounds of the replicas are not separated.

Validation is done by inspection of schematic and PCB layout, and the electrical measurements on the test item MAV2.

[HW-IRS_PAM_80][HW-IRS_PAM_116][HW-IRS_PAM_83]

[FVTSR_PAM_0030]

Test case: **FVTR_FKT_18**

Test: **Input / Output circuits' protection**

Description: Input and Output circuits of the module shall have protection against transients, overvoltage, reverse polarity, overload and short circuit.

Verification object: [CD-PAM]

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Failure criteria	Input/ Output circuits do not have a protection as described above.
Date:	Aug. 02, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	1. Both Input and output circuits for PWM are protected as follows:

- Against transients, over-voltage and reversed polarity at the PWR Input and PWM Output by TVS CR306, CR307 type SMCJ150 and diode CR305 type STTH1R02;
- Against short circuit in PWM Output by fast electronic protection self-resettable circuit, response time is about 4 µs and 2.5A fuse F301.

2. Output circuits for Analogue Current and Voltage Output are protected as follows:

- Against transients, over-voltage and reversed polarity by TVS CR402 type SM15T30A and diode CR403 type STTH1R02;
- Against short circuit and open circuit by fast electronic protection circuit built into the Analogue Driver MX403 type AD5751 and 0.14A resettable fuse F401.

[HW-IRS_PAM_67][HW-IRS_PAM_66]

[FVTSR_PAM_0031]

Test case:	**FVTR_FKT_19**
Test:	**Galvanic isolation**
Description:	The module shall provide galvanic isolation for the external I/O signals, as defined in TOP Insulation Coordination document [INSUCO_TOP].
Verification object:	[CD-PAM]
Failure criteria	No galvanic isolation is provided as above.
Date:	Aug.02, 2018

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Test carried out by: Andy Feng, HWCC

Test verified by: Michael Sacson, HWCC

Result: **Passed**

Comment: See FVTR_FKT_02, FVTR_HVT_01

[HW-IRS_PAM_65][HW-IRS_PAM_114][HW-IRS_PAM_227]

[FVTSR_PAM_0032]

Test case: **FVTR_FKT_20**

Test: **Hot-Swap**

Test Item MAV6, MAV7

Description: The module shall be pluggable and removable under voltage without causing any damage.

Failure criteria No hot-swap is possible for the PAM module.

Date: Aug.02, 2018

Test carried out by: Andy Feng, HWCC

Test verified by: Michael Sacson, HWCC

Result: **Passed**

Comment: 1. 24V power rail will be served by the PAM board while the 5V power rail will be served by the I/O Processor board.

2. Test requires Subrack and other cards. Test has been performed at SW simulator Test Subrack.

3. Test also has been performed during TOP System Integration, see [ITR_PRD_TOP].

[HW-IRS_PAM_265]

[FVTSR_PAM_0033]

Test case: **FVTR_FKT_21**

Test: **Storage device**

Test Item MAV8

Description: The PAM board shall provide two programmable serial storage devices of 1 Mbit (one for each replica) for the module type ID

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	information with an I2C interface to I/O Processor daughterboard. The function shall provide a 16 bit encoded data for identifying the board assembly type protected using a 16 bit CRC.
Failure criteria	No storage device provided as above.
Date:	Aug.02, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	An I2C EEPROM is provided on each channel. Memory has been tested by writing, reading and verifying a pattern by using KETEREX USB-910H Embedded System Interface.

[HW-IRS_PAM_291][HW-IRS_PAM_68]

[FVTSR_PAM_0034]

Test case:	**FVTR_FKT_22**
Test:	**Reduced effects of conducted interference**
Description:	The PAM design shall reduce effects of conducted interference from train lines and power lines to internal circuits by minimizing the capacitive and inductive coupling on the PCB.
Verification object:	[PBATD_PAM]
Failure criteria	The layout does not follow the above requirement.
Date:	Aug.02, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	The 10-layer PCB has been designed in provision to reduce effects of conducted interference from train lines and power lines to internal circuits by minimizing the capacitive and inductive coupling on the PCB. The PAM board assembly passed the EMC test at the Subrack level, see [QTR-PRD-HW_COMM-IO_SR].

[HW-IRS_PAM_82]

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[FVTSR_PAM_0035]

Test case:	**FVTR_FKT_23**
Test:	**Common ground of the replicas to chassis isolation**
Description:	The PAM board shall be designed with signal ground isolated from chassis (500 Vrms)
Verification object:	[PBATD_PAM], MAV1
Failure criteria	Common ground not isolated from the chassis or the isolation is lower than 500 Vrms.
Date:	October 2, 2013
Test carried out by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	See FVTR_HVT_01

[HW-IRS_PAM_81][HW-IRS_PAM_114][HW-IRS_PAM_227]

[FVTSR_PAM_0036]

Test case:	**FVTR_FKT_24**
Test:	**Front panel LED Interface**
Test Item	MAV8
Description:	The module shall provide two dual LEDs on the faceplate, one for each replica. The dual LEDs shall be green/red. Each LED shall be controlled by its corresponding replica, according to the I/O microcontroller daughterboard signal assignments shown in [HWADD_TOP].
Failure criteria	PAM statuses LEDs are not realized as above.
Date:	Aug. 02, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	1. LEDs are present and LEDs are new type LED recently updated. 2. LEDs lit correctly when driven with 3.3V. 3. LEDs lit correctly at SW simulator Test Subrack.

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[HW-IRS_PAM_69][HW-IRS_PAM_219]

[FVTSR_PAM_0037]

Test case:	**FVTR_FKT_25**
Test:	**Exposed metal surfaces**
Description:	Exposed metal surfaces shall be smoothed or de-burred to prevent injury to personnel due to contact.
Test Item	MAV8
Failure criteria	Exposed metal surfaces are not treated as above.
Date:	Aug. 02, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	Exposed metal surfaces shall be smoothed.

[HW-IRS_PAM_288]

[FVTSR_PAM_0038]

Test case:	**FVTR_FKT_26**
Test:	**Power and current consumption**
Test Item	MAV1
Description:	The PAM circuits shall be internally powered from three separate power rails: 3.3VDC, 5VDC and 24VDC, supplied from CAN I/O daughterboard. The maximum current drawn from each 3.3VDC power rail shall not exceed 200mA. The maximum current drawn from each 5VDC power rail shall not exceed 200mA. The maximum current drawn from each 24VDC power rail shall not exceed 150mA. The supply voltages generated on the PAM board and used for input circuits shall be supervised. The PAM shall provide protection for input power rails against over-load and noise generated inside the PAM.

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Failure criteria The current consumption rises above the limits specified or the power lines requirements are not satisfied.

Date: September 4, 2014

Test carried out by: Michael Schueler, HWCC

Result: **Passed**

Comment: schematic reading: PAM mainboard is supplied by two separate power rails for each supply voltage (3.3V, 5V, 24V), supplied from CAN I/O daughterboard
schematic reading: Each power rail is noise filtered and overload protected: 5V rail and 3.3V rail overload protected by Microcontroller Daughterboard, 24V rail overload protected by PAM main board

Measured current consumption is shown the Table 14 below.

Current consumption from 24V rail in Replica 1 comparing to Replica 2 is higher, because both Disconnect Relays are supplied from 24V rail of Replica 1.

Table 15: Current Consumption from Power Rails

Replica	Power Supply [v]	Max. Current [mA]	Measured Current [mA]
R1	3.3	200	5.4
	5.0	200	102,4
	24.0	150	80.1
R2	3.3	200	9.3
	5.0	200	98.8
	24.0	150	29.8

[HW-IRS_PAM_236][HW-IRS_PAM_237][HW-IRS_PAM_268][HW-IRS_PAM_238][HW-IRS_PAM_240]

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7 ISOLATION – HIGH VOLTAGE TEST

[FVTSR_PAM_0039]

Test case: **FVTR_HVT_01**

Test: **High voltage**

Test Item: MAV4

Description: The electric strength between the net classes has to be tested with the voltages according to the values declared in [INSUCO_TOP]. The access to the circuit has to be realized via the connectors.
Note: 500Vdc insulation resistance test shall be carried out before and after each high voltage test and the values recorded.

Table 16: High voltage circuits definition

Signal Group	Connector/Part	Pins	Comment
FGND	Front plate	N/A	Chassis
C6.1_I_IN	Back connector P3	B14, D14	Analogue Current Input
C6.1_V_IN	Back connector P3	B20, D20	Analogue Voltage Input
C6.2_1	Back connector P3	B2, D2, D4	PWM1 Output
C6.2_2	Back connector P3	B30, D30, D32	PWM2 Output, not populated for 7HA02944BAAA variant
C6.2_3	Back connector P3	B8, B10, D8, D10	Analogue Output 1
C6.2_4	Back connector P3	B24, B26, D24, D26	Analogue Output 2, not populated for 7HA02944BAAA variant
C4.1	Connector P1	All pins	Replica 1 Interface
C4.2	Connector P2	All pins	Replica 2 Interface

Comment: Replica 1 and Replica 2 have common GND.

The test voltages defined in Table 14 shall be applied for 60 seconds. The current must not exceed the specified current (1.5 mA for each 1000 Vrms test voltage, or equivalent 1414 Vdc).

Table 17: High voltage test conditions

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From	To	Test Voltage	Test Time, (s)	Maximum Leakage, (mA)
FGND	C4.1, C4.2	500 Vrms	60	N/A*
FGND	C6.1_I_IN, C6.1_V_IN, C6.2_1, C6.2_2, C6.2_3, C6.2_4	±2828 Vdc	60	3.0
C4.1, C4.2	C6.1_I_IN, C6.1_V_IN, C6.2_1, C6.2_2, C6.2_3, C6.2_4	±2828 Vdc	60	3.0
C6.2_1	C6.2_2, C6.2_3, C6.2_4	±1414 Vdc	60	1.5
C6.2_2	C6.2_3, C6.2_4	±1414 Vdc	60	1.5
C6.2_3	C6.2_4	±1414 Vdc	60	1.5

* No collapse of the applied voltage shall occur.

Failure criteria: The test is failed, if one of the following conditions occurs:

- The breakdown or flashover occurs;
- The current exceeds 1.5 mA per 1000 Vrms test voltage (3 mA for 2000 Vrms test, 0.75 mA for 500 Vrms tests);
- Fundamental deviation from the initial measurements, minimum acceptable value 20 MΩ as per [TEST], is noticed during the second insulation measurement.

Date: October 2, 2013

Test carried out by: Michael Sacson, HWCC

Result: **Passed**

Comment:
1. No breakdown or flashover occurs
2. No fundamental deviation from the initial measurements is noticed during the second insulation measurement.
3. Due to using DC/DC converters rated at 3000 VDC, Test steps from 2 to 6 have been performed at equivalent DC voltage, for both polarities.

The High Voltage test results shown in Table below:

Table 18: High Voltage Test Results

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Step	From	To	Test Voltage	Test Time, (s)	Result
1	FGND	C4.1, C4.2	500 Vrms	60	Pass, 0mA
2	FGND	C6.1_I_IN, C6.1_V_IN, C6.2_1, C6.2_2, C6.2_3, C6.2_4	±2828 Vdc	60	Pass, < 0.1mA
3	C4.1, C4.2	C6.1_I_IN, C6.1_V_IN, C6.2_1, C6.2_2, C6.2_3, C6.2_4	±2828 Vdc	60	Pass, < 0.1mA
4	C6.2_1	C6.2_2, C6.2_3, C6.2_4	±1414 Vdc	60	Pass, < 0.1mA
5	C6.2_2	C6.2_3, C6.2_4	±1414 Vdc	60	Pass, < 0.1mA
6	C6.2_3	C6.2_4	±1414 Vdc	60	Pass, < 0.1mA

[HW-IRS_PAM_227][HW-IRS_PAM_114]

8 INDUSTRIALIZATION AND ECOLOGICAL

[FVTSR_PAM_0040]

Test case: **FVTR_IND_01**

Test: **Materials used on the board**

Description: Selection of components and materials shall follow fire protection guidelines (Zero halogen, Non-PVC materials).

Verification object: [EBOM_PAM]

Failure criteria: Materials used are violating the rules above.

Date: Aug. 02, 2018

Test carried out by: Andy Feng, HWCC

Test verified by: Michael Sacson, HWCC

Result: **Passed**

Comment: All components and materials used for PAM assembly are approved for use in Railway Signalling Division and comply with fire protection guidelines, no warnings in TCIS and PALMA.

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[HW-IRS_PAM_242]

9 RELIABILITY

[FVTSR_PAM_0041]

Test case:	**FVTR_REL_01**
Test:	**Reliability**
Description:	Mean Time between Failures (MTBF) of the unit shall be calculated in conformance with AlcaRel failure rates (Alcatel Common Method according RAM plan). The module shall have MTBF of greater than 100,000 hours.
Test equipment:	AlcaRel Software
Failure criteria:	MTBF is not met, or the MTBF is not calculated with the above tools.
Date:	August 18, 2014
Test carried out by:	Michael. Schueler, HWCC
Result:	**Passed**
Comment:	678656 hours (77 years) in Ground Fixed environment at 45ºC for 7HA02944AAAA variant, see [RR-PBA_PAM]

[HW-IRS_PAM_233][HW-IRS_PAM_234]

[FVTSR_PAM_0042]

Test case:	**FVTR_REL_02**
Test:	**Operational life time**
Description:	The module shall be designed for a design life of at least 20 years.
Verification object:	[RR-PBA_PAM]
Failure criteria:	The expected operational life is less than 20 years.
Date:	August 18, 2014
Test carried out by:	Michael. Schueler, HWCC
Result:	**Passed**
Comment:	The MTBF is calculated and equivalent to 36 years.

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[HW-IRS_PAM_231]

[FVTSR_PAM_0043]

Test case:	**FVTR_REL_03**
Test:	**Mean Time to Repair**
Description:	The PAM module shall have a mean time to repair less than 30 minutes.
Verification object:	[HW-ID_PAM], MAV2, MAV3
Failure criteria:	The mean repair time exceeds 30 minutes.
Date:	September 5, 2014
Test carried out by:	Michael Schueler, HWCC
Result:	**Passed**
Comment:	Assuming repair is to remove and replace the board assembly.

[HW-IRS_PAM_232]

10 SAFETY REQUIREMENTS

[FVTSR_PAM_0044]

Test case:	**FVTR_SAF_01**
Test:	**Safety standards**
Description:	The design shall conform to the applicable safety standards defined in [HWADD_TOP]
Failure criteria:	The design does not meet Safety standards specified in [HWADD_TOP].
Test carried out by:	SAF
Result:	**Passed**
Comment:	Safety was involved in the design reviews and approved the design from safety aspects. Further details should be provided by Safety Experts from Safety department.

[HW-IRS_PAM_223]

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[FVTSR_PAM_0046]

Test case:	**FVTR_SAF_03**
Test:	**Independence of the replicas**
Description:	Circuits, which belong to one I/O channel (Replica 1), shall be independent from the other channel (Replica 2) by physical channel separation such that no single hardware fault can be common to both replicas or unsafely affect both channels.
Verification object:	[PBATD_PAM], [CD_PAM].
Failure criteria:	The above requirement is not satisfied.
Date:	August 18, 2014
Test carried out by:	Michael Schueler, HWCC
Result:	**Passed**
Comment:	Safety distance is implemented between channels in CAD constraints. Safety resistors [SAC] have been used for replica's separation.

[HW-IRS_PAM_225]

[FVTSR_PAM_0047]

Test case:	**FVTR_SAF_04**
Test:	**Development process**
Description:	Development process shall comply with Safety Integrity Level 2 as defined in [HWADD_TOP].
Failure criteria:	The above requirement is not satisfied.
Date:	August 18, 2014
Test carried out by:	Michael Schueler, HWCC
Result:	**Passed**
Comment:	Development process bases on PLC Ed. 19.

[HW-IRS_PAM_224]

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[FVTSR_PAM_0052]

Test case:	**FVTR_SAF_05**
Test:	**Surge protection devices**
Description:	Surge protection devices shall not ignite or melt, nor cause ignition of adjacent surfaces, when used within rated operating or surge conditions.
Verification object:	[EBOM_PAM]
Failure criteria:	Surge protection devices do not comply with requirement shown above.
Date:	August 18, 2014
Test carried out by:	Michael Schueler, HWCC
Result:	**Passed**
Comment:	Specifications for Surge Protection devices (TVS) has been analysed. TVS will not ignite or melt, nor cause ignition of adjacent surfaces, when used within rated operating or surge conditions.

[HW-IRS_PAM_221]

[FVTSR_PAM_0053]

Test case:	**FVTR_SAF_06**
Test:	**Fire protection compliance**
Description:	The module shall use components and materials which are compliant with the fire protection guidelines specified in the following standards: [EN 45545-2] [EN 45545-5] [NF F16-101] [NF F16-102] [LUL 2-01001-002]
Verification object:	[EBOM_PAM]
Failure criteria:	The above requirement is not satisfied.

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Date: September 9, 2014

Test carried out by: Michael Schueler, HWCC

Result: **Passed**

Comment:

[HW-IRS_PAM_289]

[FVTSR_PAM_0054]

Test case: **FVTR_SAF_07**

Test: **Restricted materials**

Description: The module shall restrict or justify the use of the following materials:

- Polyvinyl chlorides
- Asbestos
- Cadmium
- Beryllium
- Cobalt
- Lead
- Arsenic
- Hexavalent chromates
- Zinc
- Bromine flame proof agents
- Mercury.

Verification object: [EBOM_PAM]

Failure criteria: The above requirement is not satisfied.

Date: September 9, 2014

Test carried out by: Michael Schueler, HWCC

Result: **Passed**

Comment: No restricted materials used in PAM design.

[HW-IRS_PAM_290]

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[FVTSR_PAM_0056]

Test case: ***S* FVTR_SAF_08**

Test: **Inherent fail-safe components SAC usage validation**

Description: All safety components shall be verified for their correct usage according to their safety application conditions.

Failure criteria: Constraints for usage are not considered.

Reference: [SAC_GUIDE], [SAC_MELF0204], [CD_PAM], [EBOM_PAM], [PBATD_PAM]

Date: September 25, 2014

Test carried out by: Michael Sacson, HWCC

Result: **Passed**

Comment:

Disconnect Relays in PAM design are used to separate PWM outputs; Output separation for PAM is not a safety function.

According to the [SAC_GUIDE] and [EBOM_PAM] analysis, the following inherent fail-safe components are used on the PAM:

Reference	Type	Value	Purpose
R22, R24	MELF 0204 1AB 13984 0010	562 Ω	Trainline protection from short circuit and replica separation
R26, R28	MELF 0204 1AB 13984 0008	362 Ω	Trainline protection from short circuit and replica separation
R31, R33, R35, R37	MELF 0204 1AB 13984 0027	15 kΩ	Replica separation
R23, R25	MELF 0204 1AB 13984 0030	27.4 kΩ	Trainline protection from short circuit and replica separation
R27, R29	MELF 0204 1AB 13984 0023	6.81 kΩ	Trainline protection from short circuit and replica separation
R46, R47, R51, R52	MELF 0204 1AB 13984 0013	1 kΩ	Replica separation

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Reference	Type	Value	Purpose
R17, R48, R55, R428_3, R428_4, R429_3, R429_4, R431_3, R431_4	MELF 0204 1AB 13984 00XX	Although these components are marked as safety components on the schematic and BOM they were picked for their power handling and stability rather than their intrinsic failsafe properties. They have no effect on safety.	

The analysis against the conditions [SAC_MELF0204] is shown below:

Requirement	Value	Required Evidence	Evidence	SAC Compliance, Yes/No
5.1.1.a) Maximum operating temperature / Permissible film temperature	+ 125 °C	*1.Check IRS, if the maximum operating temperature fit to the required temperature range of the application (board / device)* *2. Show with temperature test of the board / device, that this condition will be fulfilled (e. g. by measurement of environmental temperature in the surrounding of the component). Consider the derating of the resistor above 70°C.*	1. [HW-IRS_PAM] states the maximum board level operating temperature is 85°C. 2. See [QTR-PRD-HW_COMM-IO_SR] Due to derating the power limit at 85°C is 175 mW.	**Yes**
5.1.1.b) Minimum operating temperature	- 55 °C	*1.Check IRS, if the minimum operating temperature fit to the required temperature range of the application (board / device)* *2. Show with temperature test of the board / device, that this condition will be fulfilled. Because of heating in operation (increase of component temperature), a measurement of environmental temperature in the surrounding of the component is not needed.*	1. [HW-IRS_PAM] states the maximum board level operating temperature is -40°C. 2. See [QTR-PRD-HW_COMM-IO_SR]	**Yes**

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5.1.1.c) Pollution degree	≤ PD2	*Check the clearance and creepage distances in the finalized layout and give an according statement.*	Creepage and clearance distances are taken from [INSUCO_TOP] which uses PD2 for external layers and PD1 for internal layers.	**Yes**
5.1.2.a) Maximum operating voltage	200 V_{eff}	*Check schematic for possible maximum voltage, also in case of error.*	R22, R24: 5.6 V R26, R28: 3.9 V R31, R33, R35, R37: 20 V R23 ,R25: 16 V R27, R29: 4 V R46, R47, R51, R52: 3.3 V	**Yes**
Requirement	**Value**	**Required Evidence**	**Evidence**	**SAC Compliance, Yes/No**
5.1.2.b) Limiting rated power P_{70}	0.25 W	*Check schematic for possible maximum power rating: consider the derating above 70°C.* *Calculate power rating with possible maximum voltage also in case of error.* *The limited power rating must never be exceeded.*	Due to derating the power limit at 85°C is 175 mW R22, R24: 56 mW R26, R28: 39 mW R31, R33, R35, R37: 27 mW R23, R25: 9 mW R27, R29: 4 mW R46, R47, R51, R52: 11 mW	**Yes**
5.1.2.c) Isolation voltage	280 V (DC or AC peak 1 min); 75 V (DC or AC continuous)	*Check of isolation voltage not possible after board/device production, therefore:* *Perform optical check of final layout, that there are no copper tracks below resistor body (between footprints).*	Checked layout. No metal under resistors: R22, R23, R24, R25, R26, R27, R28, R29, R31, R33, R35, R37, R46, R47, R51, R52	**Yes**
5.1.2.d) Clearance and creepage	On the assembled board according to Mü 8004, Teil 37200	*For the maximum operating voltage of 200 V_{eff} (according to 5.1.2.a) and <= PD2 the clearance and creepage distance is according to Mü 8004, Teil 37200 always fulfilled, if the over voltage in the circuit is lower than 1.5 kV impulse voltage (impulse voltage type 1.2/50 µs).*	The impulse over voltage U_{Ni} on the board for each resistor is below 1.5 kV.	**Yes**

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All the SACs for the safety components are fulfilled.

[HW-IRS_PAM_226]

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11 QUALIFICATION, TESTING AND MAINTENANCE

[FVTSR_PAM_0048]

Test case:	**FVTR_QTM_01**
Test:	**Environmental standards for qualification**
Description:	The environmental qualification of the module shall be performed in accordance with CENELEC and AREMA.
Failure criteria:	The qualification does not comply with the above standards.
Date:	August 18, 2014
Test carried out by:	Michael Schueler, HWCC
Result:	**Passed**
Comment:	Passed by analysis: design is done in provision to comply environmental standards requirements. Actual environmental qualification will be done at Subrack or System level in accordance with CENELEC and AREMA requirements see Chapter 12.

[HW-IRS_PAM_246]

[FVTSR_PAM_0049]

Test case:	**FVTR_QTM_02**
Test:	**Functional qualification procedures**
Description:	Functional qualification of the PAM Board shall be performed according to Thales qualification procedures.
Failure criteria:	The above procedure are not followed
Date:	September 11, 2014
Test carried out by:	Michael Schueler, HWCC
Result:	**Passed**
Comment:	Functional qualification of the PAM Board has been performed according to Thales qualification procedures.

[HW-IRS_PAM_247]

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[FVTSR_PAM_0050]

Test case:	**FVTR_QTM_03**
Test:	**Diagnosis and Maintenance**
Description:	The board shall be constructed such that maintenance personnel are able to perform first order maintenance using common tools.
Failure criteria:	Special tools are needed for first order maintenance.
Date:	August 18, 2014
Test carried out by:	Michael Schueler, HWCC
Result:	**Passed**
Comment:	PAM can be maintained with common tools.

[HW-IRS_PAM_249]

[FVTSR_PAM_0057]

Test case:	**FVTR_QTM_04**
Test:	**Pollution degree compliance**
Description:	The module shall be designed and manufactured to meet pollution requirements defined in [HWADD_TOP].
Failure criteria:	Design is not complined with pollution protection requirements defined in [HWADD_TOP].
Date:	January 28, 2021
Test carried out by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	The board design (creepage and clearance) has been done as per requirements of [HWADD_TOP] and [INSUCO_TOP]: Standard EN 50124-1, Pollution Degree PD2 for operating without a cabinet. The PBA has been conformal coated as per [PBATD_PAM] requirements.

[HW-IRS_PAM_126]

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12 DOCUMENTATION

[FVTSR_PAM_0051]

Test case:	**FVTR_DOC_01**
Test:	**Documentation**
Description:	Module documentation shall be prepared according with the TOP Documentation Plan [DOP].
Failure criteria:	Documentation is not prepared according to [DOP].
Date:	September 11, 2014
Test carried out by:	Michael Schueler, HWCC
Result:	**Passed**
Comment:	All documentation is prepared according to [DOP], reviewed and released in CMCS, archived in PALMA.

[HW-IRS_PAM_251]

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13 DEFECT TREATMENT

Test case:	**TASCM00231894**
Test:	**7HA02922AAAA PAM: Mechanic**
Description:	Placement of POS 4 corrected. Hidden type label from the board assembly only -> accepted because normal behavior. Solder side cover exchanged to 3JA 69041 0075
Failure criteria:	defect not corrected
Date:	September 19, 2014
Test carried out by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	Changes have been implemented, **TASCM00231894 Status is Solved.**

Test case:	**TASCM00234614**
Test:	**7HA02944AAAA PAM Main PBA update after first prototype testing**
Description:	The following issues shall be fixed: 1. Status contacts for disconnect relays K1_3/4 and K2_3/4 shall be swapped in the Circuit Diagram and the Layout; 2. Current Sensor R430 for Analogue Output shall be moved to the low side of the Analog Output Driver MAX403 (between AN_OUT_RTN and AGND_ISO to protect the Status ADC input against High Common Mode input signal; 3. Based on the issue 2 stated above, separate returns for Analogue Output Signal and Analogue Output Power shall be created: - Pin D10 shall be re-named as AN_OUT1_PWR_RTN and disconnected from B10;

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- Pin D26 shall be re-named as AN_OUT2_PWR_RTN and disconnected from B26.

After this change:

- AN_OUT1/AN_OUT1_RTN is B8 / B10, and
- AN_OUT2/AN_OUT2_RTN is B24/ B26.

HW-IRS_PAM and HWADD_TOP shall be updated accordingly;

4. I2C address for ADC MX1 (MX2) shall be updated from 1001000 to 1001010 by re-moving Resistor R56 (R57) and re-connecting MX1 (MX2) Pin 1{ADDR} to Pin 9 {SDA}.

5. To limit and align the brightness of the LED indicators the series resistors R1, R2, R3 and R4 have to be changed to 681Ohm.

6. Replace resistors R58, R59, R62, R63 475R/1206 with 365R/1206, BCN:99220820 to protect MX1, MX2 against over-voltage after DC/DC.

7. In the PCB Layout, move location of R3, R4 far away from the Chassis (Front Plate mounting part).

8. Update both PWM Outputs to improve short Circuit Protection feature:

- Change C310 value from 100nF to 22nF to fix ringing on the rising edge;

- Change C313 value from 4.7uF to 1uFto accelerate re-charging (to be ready to generate next pulse in case of permanent short circuit event);

- Change R313 value from 10k to 3.01k to decrease sensitivity of MonoStable;

- Add resistor 100R in series with MA303 pin 7 to limit the optocoupler output current;

- Change R301 value from 10k to 30.1k;

- Place capacitor 47nF in parallel to R301 to create time delay sufficient for MonoStable re-charging.

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	9. Remove R106_1, R406_2 from the BOM: component is not populated.
Failure criteria:	defect not corrected
Date:	September 19, 2014
Test carried out by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	Changes have been implemented, **TASCM00234614 Status is Solved.**

Test case:	**TASCM00236797**
Test:	**Insertion of boards (DIM, DOM-B, DOM-V, PAM, TMM) into the TOP subracks is excessively tight**
Description:	Measurements on the actual boards, show approximately 0,6mm misalignment.
Failure criteria:	defect not corrected
Date:	September 19, 2014
Test carried out by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	Changes have been implemented, **TASCM00236797 Status is Solved.**

Test case:	**TASCM00236800**
Test:	**All TOP boards: Rating of fuses**
Description:	TOP project has received a request to verify types and ratings of all fuses used on our boards, due to reports of field failures causing reliability issues of Thales products. We must review and verify if all fuses are: - slow-blow type (time lag); - current (A) values are correctly selected with sufficient margin (typically rated at least 4 times higher than the maximum expected current);

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Failure criteria: defect not corrected

Date: September 19, 2014

Test carried out by: Michael Sacson, HWCC

Result: **Passed**

Comment: No actions required, see TASCM00236994; **TASCM00236800**

Status is Solved.

Test case: **TASCM00240677**

Test: **7HA02944AAAA PAM Main PBA update**

Description: The following issues shall be fixed:

1. I2C pull-up Resistors R64, R65, R66, R67, R405_3, R405_4, R409_3, R409_4: value shall be changed from 5.11K to 1.78K (BCN: 99216901) to improve signal shape quality;

2. Disconnect Relays Status replica-separation safety resistors R46, R47, R51, R52: value shall be changed from 10K to 1K (BCN: 1AB139840013) to ensure correct value of logic LOW signal when relays are energized.

Failure criteria: defect not corrected

Date: September 19, 2014

Test carried out by: Michael Sacson, HWCC

Result: **Passed**

Comment: Changes have been implemented, **TASCM00240677**

Status is Solved.

Test case: **TASCM00246198**

Test: **PAM generating too much heat**

Description: The recent temperature tests revealed certain areas of the PAM is measured to be +22 C above ambient. In particular, the R2 disconnect relay located between the sandwiched area of the DB and MB.

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An investigation needs to be performed to determine the best way forward to decrease the amount of heat generated.

Failure criteria: defect not corrected

Date: September 19, 2014

Test carried out by: Michael Sacson, HWCC

Result: **Passed**

Comment: **Status of the TASCM00246198 is solved.**

Test case: **TASCM00249038**

Test: **7HA02944AAAA PAM update as per Bug Fix Cycle functional verification**

Description: 1. Diodes CR302_1 and CR302_2 shall be removed, due to signal PWM_DISABLE is not used in existing board variants and this function is not required by [HW-IRS_PAM]. In case of erroneous signal on not controlled FPGA output, the presence of these diodes may create PWM function interruption. EBOM and Electrical Schematic shall be updated.
2. The power pin 8 of optocouplers MA303_1 and MA303_2 has been re-connected from 5V power rail to 3V3 power rail by fly wires. The reason is: PWM_CB signal supplied from the optocoupler output to FPGA input shall be 3.6V max; this input is not 5V tolerable, and FPGA may be damaged. Electrical Schematic and existing prototypes shall be updated.

Failure criteria: defect not corrected

Date: September 19, 2014

Test carried out by: Michael Sacson, HWCC

Result: **Passed**

Comment: **I**tem/issue 1 is not required by architecture/specification/ to limit design risk; see TASCM00274405 in CQ. Item/issue 2 is implemented, **TASCM00249038 Status is Solved**.

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Test case:	**TASCM00260142**
Test:	**7HA02944xAAA PAM - no test access to the signal EGND from bottom site**
Description:	The resistor R15 cannot be measured by Flying Probe Test, because we have no test access to the signal EGND from bottom site.
Failure criteria:	defect not corrected
Date:	September 19, 2014
Test carried out by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	Changes have been implemented, **TASCM00260142 Status is Solved.**

Test case:	**TASCM00262317**
Test:	**Add termination resistor to PAM for PWM output**
Description:	Need termination resistor in PAM board for PWM output. Solution needs to be defined as a big resistor for high current will not fit on the board.
Failure criteria:	defect not corrected
Date:	September 19, 2014
Test carried out by:	Michael Sacson, HWCC
Result:	**Not applicable**
Comment:	Not applicable for PAM, shall be done at sub-rack level. **Ticket TASCM00262317 has been Rejected.**

Test case:	**TASCM00269270**
Test:	**PWM output from PAM board is not correctly shutoff**
Description:	PWM output from PAM board is not correctly shutoff when the output is turned off when the PWM output is high.
Failure criteria:	defect not corrected
Date:	September 19, 2014
Test carried out by:	Michael Sacson, HWCC

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Result:	**Not applicable**
Comment:	Not a bug in HW; SW to be changed. **Ticket TASCM00269270 has been Rejected.**

Test case:	**TASCM00272478**
Test:	**Connector from Mainboard to Daughterboard to be changed**
Description:	The connector at the mainboards for DIM, DOM, TMM and PAM need to be changed because of manufacturing issues. The exiting connectors do not have locking pins. This leads to the problem, so that the connectors can move during soldering. This leads to not fitting daughter-boards. The existing part-number 99235240 must be exchanged with 99241476.
Failure criteria:	defect not corrected
Date:	September 19, 2014
Test carried out by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	Changes have been implemented, **TASCM00272478 Status is Solved.**

Test case:	**TASCM00307459**
Test:	**Some components foot print is not correct**
Description:	The components Pos. 1302 (MA302_1 / MA302_1 /) have the false footprint in the ODB-data.
Failure criteria:	defect not corrected
Date:	Aug 13, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	Changes have been implemented, **TASCM00307459 Status is Solved.**

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Test case:	**TASCM00368348**
Test:	**96V was not considered for variant AAAA**
Description:	PAM 7HA-02944-AAAA was designed for 110V power rail and its input voltage range doesn't cover the whole voltage range for 96V, which is stated in the standard EN 50155.
Failure criteria:	**defect not corrected**
Date:	Aug 13, 2018
Test carried out by:	Andy Feng, HWCC
Test verified by:	Michael Sacson, HWCC
Result:	**Passed**
Comment:	Changes have been implemented, **TASCM00368348 Status is Solved.** New variant 7HA-02944-AAAB was designed to support PWM voltage 96V and 110V, which will eventually be used to replace 7HA-02944-AAAA after validation.

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14 ENVIRONMENTAL AND EMC

Environmental and EMC requirements have been analysed during detailed design stage and FVTS/R process to ensure design is done in provision to meet conditions as listed in [HWADD_TOP].

Environmental and EMC requirements have been tested at the Subrack or System Level and are covered in [QTR-PRD-HW_COMM-IO _SR].

Reference: [HW-IRS_PAM_63],[HW-IRS_PAM_105],[HW-IRS_PAM_106], [HW-IRS_PAM_108],[HW-IRS_PAM_110],[HW-IRS_PAM_112], [HW-IRS_PAM_118],[HW-IRS_PAM_120],[HW-IRS_PAM_122], [HW-IRS_PAM_124]

END OF DOCUMENT

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